

FAB / METROLOGY / QUALIFICATION / FOUNDRY

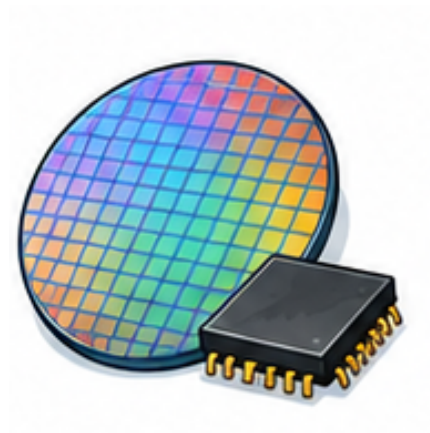
Semiconductor English

Name the layer. Qualify the result.
Align the handoff.

Phil Stilwell

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Eight scenarios. Eight extended conversations.

INSIDE THE BOOK

192 vocabulary entries / 128 reusable phrases
160 dialogue turns / 80 numbered dialogue gaps
Word banks and explained answer keys

Upper-intermediate to advanced (B2-C1). For process, integration, equipment, yield, product, test, and foundry-facing teams.

2026 edition

Eight scenarios. Eight skills.

Eight semiconductor conversations connecting process context, measurement, yield, qualification, and delivery commitments.

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02 Lithography, Reticles, and Critical Dimensions

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One completed test is not complete qualification | Explain qualification status test by test, preserve configuration scope, and correct an overbroad customer claim.

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A number needs its process context.

Semiconductor discussions compress complex evidence into phrases such as CD shift, yield gain, matched tool, and tape-out ready. Useful communication restores the missing context: layer, stage, lot, measurement recipe, test coverage, or decision status. These cases practice precise technical questions and concise updates that help different specialties work from the same facts.

Locate the observation

Identify the layer, process stage, lot or wafer, tool, and time relevant to the claim. A change after etch is not automatically the same as a change measured in developed resist.

"The reported shift is on M2 after etch; the measurement recipe also changed."

Keep the denominator visible

Yield and defect figures depend on what was counted, sampled, and compared. State the population and measurement basis before interpreting the difference.

"The pilot result is 92 passing die out of 100 tested, with a different product mix."

Separate settings from performance

Identical recipe names do not prove tools behave alike. Compare the evidence against a defined matching criterion and distinguish availability from qualification.

"Both means are within specification, but the pair misses our matching limit."

Do not compress pending into complete

A finished test, a submitted layout, and a possible capacity slot each have a limited meaning. Name the remaining checks and who can authorize the next step.

"One qualification test passed; two required tests remain open."

PROFESSIONAL BOUNDARIES

Original fictional English practice, not a fabrication recipe, equipment procedure, product qualification, design sign-off, or production authorization. Numbers, lots, organizations, and schedules are invented. Real work must follow approved process documents, safety rules, applicable standards, customer requirements, and authorized engineering decisions.

Back-end does not always mean packaging

A Read the situation

Explain the current process stage and ownership without confusing wafer interconnect fabrication with package assembly.

New product coordinator Eric describes lot L24 as being in packaging because the status says BEOL. Integration engineer Min explains that this lot is still in wafer fabrication, building on-wafer interconnect layers. Its current operation concerns metal layer M2. In this fictional route, an inspection review must be completed before the next authorized operation. Packaging has not started, and the lot is on hold pending that review. Eric must correct the customer update without inventing a shipping date.

PEOPLE / Eric: Product coordinator | Min: Process integration engineer

EXPAND THE ABBREVIATION AT A HANDOFF

Back-end can refer to different stages in different teams. Use the full process name when crossing organizational boundaries, then identify the actual operation and lot status. Familiar shorthand is efficient only when the receiver shares its meaning.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Where is L24 now?

- A In completed package assembly
- C In customer field qualification

- B In final shipment inspection
- D In wafer fabrication at an M2-related operation

2. What does BEOL mean in this case?

- A Back end of line, forming on-wafer interconnects
- C A confirmed customer delivery date

- B Completed assembly and final test
- D A shipping release status

3. What prevents the next operation?

- A A customer order cancellation
- C A pending inspection review and lot hold

- B A recorded shipment approval
- D A completed packaging qualification

[Answers and explanations: page 76.](#)

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

wafer

A semiconductor substrate on which many devices are fabricated.

IN USE process a wafer

die

An individual circuit area or chip separated from a wafer.

IN USE identify the die

fab

A semiconductor fabrication facility.

IN USE move material through the fab

wafer fabrication

Processes that form devices and interconnects on wafers.

IN USE track wafer fabrication

process flow

The defined sequence of manufacturing operations.

IN USE review the process flow

process integration

Coordination of process steps into a functioning device flow.

IN USE assess process integration

FEOL

Front end of line, principally the formation of active devices.

IN USE complete FEOL processing

MOL

Middle of line, connecting devices to the interconnect structure.

IN USE review MOL interfaces

BEOL

Back end of line, forming the on-wafer interconnect structure.

IN USE track BEOL progress

interconnect

A conductive structure connecting circuit elements.

IN USE form an interconnect

metal layer

A defined level of conductive routing in a chip.

IN USE identify the metal layer

via

A conductive connection between interconnect levels.

IN USE inspect a via

Precision check

BEOL is still part of wafer fabrication in this conversation. It is not interchangeable with package assembly, although some organizations also use back-end for assembly and test. Expand the term and state the actual operation before communicating status.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

dielectric

An electrically insulating material used between conductors.

IN USE deposit a dielectric

process module

A related group of operations within the fabrication flow.

IN USE qualify a process module

lot

A tracked group of wafers or units processed under defined identity.

IN USE identify the lot

wafer ID

The identifier assigned to an individual wafer.

IN USE confirm the wafer ID

route

The controlled sequence assigned to a tracked product or lot.

IN USE follow the approved route

traveler

A record accompanying material through its required operations.

IN USE update the traveler

WIP

Work in process: material not yet through the defined manufacturing flow.

IN USE report WIP status

lot hold

A restriction preventing a lot from proceeding without disposition.

IN USE place a lot on hold

disposition

An authorized decision about the next treatment or status of material.

IN USE obtain lot disposition

wafer sort

Electrical testing of individual die while on the wafer.

IN USE schedule wafer sort

dicing

Separating individual die from a wafer.

IN USE complete wafer dicing

package assembly

Integrating and protecting die in the intended package structure.

IN USE begin package assembly

Precision check

A lot at a later fabrication stage is not necessarily ready to ship. L24 remains on hold for inspection review. Its next move depends on the approved route and authorized disposition; no packaging start or delivery commitment is supplied.

Phrases you can actually use

C Notice the language / 1 of 2

CLARIFY THE ABBREVIATION

"Here BEOL means back end of line."

NAME THE LAYER

"The current operation concerns metal layer M2."

EXPLAIN INTEGRATION

"We need to preserve compatibility between successive process modules."

CHECK THE ROUTE

"What is the next operation on the approved route?"

LOCATE THE WORK

"The lot is still in wafer fabrication."

SEPARATE THE STAGES

"On-wafer interconnect formation is not package assembly."

IDENTIFY THE MATERIAL

"Which lot and wafer IDs does the update cover?"

STATE THE HOLD

"L24 is on hold pending inspection review."

Back-end

May mean BEOL or assembly and test; clarify context.

Complete

Specify the operation and whether inspection or release is included.

Lot

A tracking group, not necessarily one die or one finished package.

Phrases you can actually use

C Notice the language / 2 of 2

KEEP AUTHORITY EXPLICIT

"The next move needs authorized disposition."

DISTINGUISH THE TESTS

"Wafer sort and final packaged-device test are different steps."

CORRECT THE UPDATE

"Packaging has not started for this lot."

ASK FOR THE STATUS BASIS

"Does completed mean processed, inspected, or released?"

AVOID A SHIPPING INFERENCE

"This stage does not establish a shipment date."

CONFIRM THE HANDOFF

"Which team owns the inspection review?"

PRESERVE TRACEABILITY

"Keep the traveler linked to the lot and operation."

CLOSE PRECISELY

"Report the stage, hold reason, owner, and next decision."

Next

Must follow the approved route and current restrictions.

Ready

Needs a named next step and its prerequisites.

Released

Means a specific authorized status change, not merely elapsed time.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which customer update is accurate?

- A** L24 remains in wafer fabrication at M2, on hold for inspection review; packaging has not started.
- B** L24 is packaged because BEOL contains the word back.
- C** L24 is ready to ship once the current meeting ends.
- D** All wafers in the fab have completed final test.

2. Why expand BEOL for the coordinator?

- A** BEOL always means package assembly in every organization.
- B** Acronyms eliminate the need to name an operation.
- C** The lot ID alone reveals its full status.
- D** Different teams can use back-end for different parts of the manufacturing chain.

3. What is the appropriate next-step wording?

- A** Promise shipment before checking the remaining flow.
- B** The lot awaits inspection review and authorized disposition under its route.
- C** Move the lot because M2 is usually near completion.
- D** Skip the review because packaging has not started.

4. Which distinction is correct?

- A** Wafer sort tests die on the wafer; package assembly integrates die into packages.
- B** Wafer sort is another name for every packaging operation.
- C** Dicing creates the entire interconnect stack.
- D** Package assembly proves all earlier fabrication stages passed inspection.

[Answers and explanations: page 76.](#)

Back-end does not always mean packaging

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

process flow / disposition / FEOL / traveler / wafer fabrication / Wafer sort / metal layer / package assembly / lot hold / BEOL

- 01 **Eric:** I wrote that L24 is in packaging because the status says BEOL. The customer wants to know whether that means finished parts will be ready soon.
- 02 **Min:** Here ¹ _____ means back end of line. We are building the interconnect structure on the wafer, so your update places the lot at a later stage than the record supports.
- 03 **Eric:** That explains the confusion. The assembly team also calls its work back-end, so I assumed both groups meant the same stage.
- 04 **Min:** Use ² _____ in the customer message, then name the actual operation. Expanding the term is especially useful when the reader does not share our process shorthand.
- 05 **Eric:** The current entry refers to M2. I should identify that layer rather than give the impression that every interconnect level or every remaining operation is complete.
- 06 **Min:** Correct. M2 is the relevant ³ _____. A layer name locates the work, but it does not by itself tell you that inspection, disposition, or later stages have finished.
- 07 **Eric:** Can you explain how that fits with the earlier device work? I need enough context to brief the customer without reciting every operation in the fab.
- 08 **Min:** ⁴ _____ principally forms the active devices, while later interconnect work connects them. The exact flow is technology-specific, so use our product route for the actual sequence and status.
- 09 **Eric:** I also see that inspection review is pending. The lot cannot simply move to its next listed operation because someone has already processed the current step.
- 10 **Min:** It is on a ⁵ _____. That restriction remains in place until the required review and authorized decision. Processed, inspected, and released are different status claims.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 10.

- 11 **Eric:** Then the update should identify what must happen before the next move, not describe the hold as if it were just a routine scheduling delay.
- 12 **Min:** Yes. We need ⁶ _____ under the approved process. The review owner can explain the decision basis; we should not predict the outcome before that evidence is assessed.
- 13 **Eric:** I will confirm the owner and preserve the lot reference. We should also keep the record connected to the individual wafers where the inspection information requires that detail.
- 14 **Min:** Keep the ⁷ _____ and relevant wafer IDs aligned with the update. Otherwise two teams may discuss different material while assuming they are referring to the same issue.
- 15 **Eric:** After fabrication, the remaining route includes other work before a finished product can ship. We should not jump directly from an interconnect entry to a delivery promise.
- 16 **Min:** Exactly. ⁸ _____, dicing, assembly, and subsequent checks have their own place in the approved route. Their timing is not established by the M2 entry alone.
- 17 **Eric:** I will correct the packaging statement now, before the customer plans around a stage the lot has not reached.
- 18 **Min:** Say that ⁹ _____ has not started for this lot. You can give a precise current status without inventing dates for the remaining operations.
- 19 **Eric:** The revised message will locate L24 in wafer fabrication at M2, explain the inspection hold, and identify the next review. It will leave shipment timing uncommitted.
- 20 **Min:** That reflects the ¹⁰ _____ accurately. The customer gets a clear explanation of progress and the pending decision instead of an optimistic interpretation of an abbreviation.

ANSWERS / EXPLANATIONS: PAGE 77

1. BEOL | 2. wafer fabrication | 3. metal layer | 4. FEOL | 5. lot hold | 6. disposition | 7. traveler | 8. Wafer sort | 9. package assembly | 10. process flow

Name the stage before the date

F Say it

Lot N6 is forming on-wafer interconnects. It is on hold for inspection review. Package assembly has not started, and no shipping date is confirmed.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

fabrication / unconfirmed / started / hold

Coordinator: "The lot is still in wafer ¹ _____."

Engineer: "Its current restriction is a lot ² _____."

Coordinator: "Package assembly has not ³ _____."

Engineer: "The shipping date remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 77

1. fabrication | 2. hold | 3. started | 4. unconfirmed

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A three-nanometer shift without a common basis

A Read the situation

Clarify a reported dimension change by identifying the layer, process stage, sampling, and measurement recipe before attributing a cause.

Lithography engineer Lina reviews a reported critical-dimension increase with metrology engineer Theo. The summary initially omits the layer and stage. The records show M2 after etch, with means of 86 nm and 89 nm. However, the two measurements used different CD-SEM recipes and different sampling sites. No comparable remeasurement, specification assessment, or cause investigation is complete. Lina must report the apparent 3 nm difference without treating it as a confirmed lithography shift or changing an exposure setting.

PEOPLE / Lina: Lithography engineer | Theo: Metrology engineer

ASK WHICH DIMENSION WAS MEASURED

CD shift sounds specific but can conceal different features, stages, or measurement settings. A useful challenge asks for the measurement basis before debating process causes. Do not assume that a post-etch result isolates lithography or that a changed mean proves a specification violation.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which layer and stage do the records identify?

- A Unspecified resist before exposure
- C M2 before every fabrication step

- B Final package thickness
- D M2 after etch

2. What is the reported numerical difference?

- A An increase of 89 nm
- C An increase of 3 nm

- B No numerical difference
- D A decrease of 3 nm

3. Why is a direct process attribution premature?

- A The layer is now identified.
- C The measurement recipes and sampling sites differ.

- B Every post-etch result can only originate in lithography.
- D Both figures use nanometers.

Answers and explanations: page 78.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

lithography

Pattern formation using exposure of a sensitive layer.

IN USE review lithography performance

reticle

A patterned optical element used in lithographic exposure.

IN USE identify the reticle

photoresist

A radiation-sensitive material used to form a pattern.

IN USE coat the photoresist

exposure dose

The delivered exposure energy per unit area.

IN USE record exposure dose

focus offset

A displacement from a defined best-focus reference.

IN USE evaluate focus offset

critical dimension

A feature dimension important to process or device performance, abbreviated CD.

IN USE measure critical dimension

CD-SEM

A scanning electron microscope used for critical-dimension measurement.

IN USE verify the CD-SEM recipe

measurement recipe

The controlled settings and analysis used to obtain a measurement.

IN USE confirm the measurement recipe

sampling site

A specified location selected for measurement.

IN USE align sampling sites

post-develop

Measured or occurring after resist development.

IN USE review post-develop CD

post-etch

Measured or occurring after an etch operation.

IN USE compare post-etch measurements

etch bias

A dimensional difference associated with pattern transfer through etch.

IN USE evaluate etch bias

Precision check

The arithmetic difference is 3 nm. That does not prove a 3 nm physical process shift because the recipes and sampled sites differ. A post-etch CD combines the history of pattern formation and transfer; it does not identify exposure as the sole cause.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

overlay

Alignment accuracy between patterned layers.

IN USE measure overlay

alignment mark

A feature used to position a layer or measurement.

IN USE inspect alignment marks

line-edge roughness

Variation of a patterned line edge from its intended shape.

IN USE assess line-edge roughness

line-width roughness

Variation in the width of a patterned line.

IN USE assess line-width roughness

pitch

The repeat distance between corresponding features.

IN USE specify feature pitch

pattern fidelity

Agreement between the intended and produced pattern.

IN USE assess pattern fidelity

optical proximity correction

Layout adjustments compensating for imaging effects, abbreviated OPC.

IN USE review optical proximity correction

focus-exposure matrix

A test pattern set spanning selected focus and dose conditions.

IN USE analyze a focus-exposure matrix

depth of focus

A usable focus range under a defined imaging criterion.

IN USE assess depth of focus

exposure latitude

A usable dose range under a defined imaging criterion.

IN USE estimate exposure latitude

measurement bias

A systematic measurement difference relative to a reference.

IN USE assess measurement bias

remeasurement

A repeated measurement under identified conditions.

IN USE request comparable remeasurement

Precision check

CD and overlay measure different properties: feature size and layer alignment. Neither should be substituted for the other. A specification judgment also needs the applicable limits and measurement basis, which are not supplied as resolved facts in this case.

Phrases you can actually use

C Notice the language / 1 of 2

LOCATE THE DIMENSION

"Which layer and feature does the CD refer to?"

STATE THE NUMBERS

"The reported means are 86 and 89 nm."

ASK ABOUT MEASUREMENT

"Did both datasets use the same CD-SEM recipe?"

AVOID A CAUSAL SHORTCUT

"Post-etch CD does not isolate exposure performance."

IDENTIFY THE STAGE

"These measurements are from M2 after etch."

QUALIFY THE DIFFERENCE

"The apparent difference is 3 nm."

CHECK SAMPLING

"Were the same types of sites sampled?"

SEPARATE METRICS

"Overlay is not another name for critical dimension."

Shift

Specify whether apparent, confirmed, or attributed.

CD

Identify feature, layer, stage, and measurement method.

Post-etch

Describes a stage, not proof of which earlier step caused a change.

Phrases you can actually use

C Notice the language / 2 of 2

REQUEST COMPARABILITY

"Remeasure on an agreed, traceable basis."

AVOID AN UNSUPPORTED ADJUSTMENT

"Do not infer an exposure change from this comparison alone."

DISTINGUISH VARIATION

"A mean shift and within-wafer variation are different questions."

DESCRIBE THE NEXT CHECK

"Resolve the measurement difference before attributing the process cause."

PRESERVE THE ORIGINALS

"Keep both original recipes with their datasets."

ASK FOR LIMITS

"Which specification applies to this feature and stage?"

CLARIFY RETICLE IDENTITY

"Which reticle revision was used for the layer?"

CLOSE THE REPORT

"State the apparent shift and the comparability limitation together."

Same recipe

Confirm the actual revision, not only the displayed name.

Out of spec

Needs the applicable limit and a valid measurement basis.

Adjust

Requires an authorized technical decision, not a language exercise inference.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which report sentence is justified?

- A** Both datasets prove the same physical result because they use nanometers.
- B** M2 is out of specification without any stated limit.
- C** The means differ by 3 nm, but recipe and site changes limit comparability.
- D** Exposure drift has been confirmed at exactly 3 nm.

2. Which comparison best helps investigate the apparent change?

- A** A packaging thickness average with no layer reference
- B** A guessed exposure correction with no review
- C** Comparable measurements on identified sites using an agreed recipe basis
- D** A different feature at an unspecified stage

3. Why is post-etch important?

- A** It makes the sampling locations irrelevant.
- B** The result follows pattern transfer and does not isolate lithography alone.
- C** It means no etch process has occurred.
- D** It establishes that every cause is in the reticle.

4. Which distinction is accurate?

- A** CD concerns feature size; overlay concerns layer alignment.
- B** CD and overlay are interchangeable labels for the same result.
- C** A reticle revision is a unit of CD.
- D** Pitch always equals every line's width.

Answers and explanations: page 78.

A three-nanometer shift without a common basis

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

pattern fidelity / Overlay / remeasurement / sampling site / reticle / critical dimension / measurement recipe / post-etch / Etch bias / measurement bias

- 01 **Lina:** The review slide says CD shifted, but it does not identify the layer or the measurement stage. I cannot tell which process team should investigate first.
- 02 **Theo:** The records identify ¹ _____ M2 measurements. The two means are eighty-six and eighty-nine nanometers, so the reported numerical increase is three nanometers.
- 03 **Lina:** That locates the result, but I still need to know whether the two measurements were obtained on a comparable basis before calling it a physical process shift.
- 04 **Theo:** The ² _____ changed between datasets. Both used CD-SEM, but sharing the instrument type does not mean their settings and analysis were identical.
- 05 **Lina:** Were the same locations measured? A different selection across the wafer could also change the average without establishing the process explanation stated in the meeting.
- 06 **Theo:** The ³ _____ selection changed too. We need the location records alongside the recipes, rather than treating the two mean values as a controlled before-and-after comparison.
- 07 **Lina:** Then I can report an apparent three-nanometer difference while making the comparability limitation explicit. I should not write that exposure drift has already been isolated.
- 08 **Theo:** Correct. The ⁴ _____ result is important, but the present comparison does not separate process behavior from measurement and sampling effects.
- 09 **Lina:** Because this is after etch, the final feature can reflect both the earlier resist pattern and the transfer step. It is not a direct exposure-only measurement.
- 10 **Theo:** Exactly. ⁵ _____ is a separate consideration in the investigation. Naming it does not establish that etch caused this case, but it prevents an unjustified shortcut to lithography.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 19.

- 11 **Lina:** Someone also asked whether the CD result proves an alignment problem. We should distinguish those metrics rather than let one technical term replace another.
- 12 **Theo:** ⁶ _____ concerns alignment between layers. It is not another name for feature width, so this CD comparison cannot be relabeled as an overlay finding.
- 13 **Lina:** Our next request should identify the relevant feature, layer, stage, locations, and agreed measurement settings. Then the engineers can compare results with a traceable basis.
- 14 **Theo:** Request comparable ⁷ _____ through the approved process. Preserve the original datasets too, because their recipes and locations are part of the explanation for what was initially reported.
- 15 **Lina:** If the difference persists, the team can investigate possible process causes with better evidence. If it does not, we still need to explain the original discrepancy.
- 16 **Theo:** Yes. Assess ⁸ _____ and other relevant effects rather than force either outcome. The aim is a supported interpretation, not a preferred conclusion that the process changed or stayed identical.
- 17 **Lina:** I will include the reticle revision in the context record as well. That gives the investigation the configuration information without implying the reticle has been found defective.
- 18 **Theo:** Good. The ⁹ _____ identity belongs with the layer history. Configuration details help evaluate hypotheses, but listing a component does not make it the cause.
- 19 **Lina:** The revised slide will keep the two means and the three-nanometer difference, state the changed measurement basis, and leave the cause and specification assessment open.
- 20 **Theo:** That protects ¹⁰ _____ discussions from unsupported conclusions. We can plan the next evidence review without treating this comparison as permission to change exposure settings.

ANSWERS / EXPLANATIONS: PAGE 79

1. post-etch | 2. measurement recipe | 3. sampling site | 4. critical dimension | 5. Etch bias | 6. Overlay | 7. remeasurement | 8. measurement bias | 9. reticle | 10. pattern fidelity

Same units, different measurement basis

F Say it

A layer has reported means of 120 and 124 nm after etch. Different measurement recipes and sampling sites were used. The cause is not established.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

4 / different / etch / unconfirmed

Engineer: "The apparent increase is ¹ _____ nm."

Reviewer: "The measurements were taken after ² _____."

Engineer: "The measurement recipes were ³ _____."

Reviewer: "The cause remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 79

1. 4 | 2. etch | 3. different | 4. unconfirmed

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A better center point is not a wider window

A Read the situation

Describe a process trial improvement without extending it to untested conditions or overlooking downstream integration effects.

Deposition engineer Yuki and integration engineer Sam review a trial film process. At the nominal setting, a defined thickness-nonuniformity metric falls from 5% to 3%; lower is better on this unchanged measurement basis. Only that center condition has been tested. The proposed low and high settings remain untested, and effects on subsequent etch and chemical mechanical planarization are not assessed. The new recipe is not released. A summary nevertheless says the complete process window is validated.

PEOPLE / Yuki: Deposition engineer | Sam: Integration engineer

CELEBRATE A BOUNDED IMPROVEMENT

An encouraging center-point result is worth reporting. State the metric and tested condition, then identify the untested range and integration questions. This lets colleagues recognize progress without confusing a promising experiment with a released process window.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What improved in the supplied trial?

- | | |
|--|---|
| A The new recipe received production release. | B The stated nonuniformity metric fell from 5% to 3% at the nominal setting. |
| C Every condition in the proposed range was tested. | D Etch and planarization were fully qualified. |

2. What is the absolute change in that percentage metric?

- | | |
|---|--|
| A A decrease of 2 percentage points | B A decrease of 2 relative percent |
| C An increase of 3 percentage points | D A decrease of 5 percentage points |

3. Which conclusion is premature?

- | | |
|---|--|
| A The low and high settings remain untested. | B Downstream integration effects remain open. |
| C The complete process window is validated. | D The nominal-setting result improved on the stated metric. |

Answers and explanations: [page 80](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

deposition

Formation of a material film on a surface.

IN USE qualify a deposition step

thin film

A material layer with a small controlled thickness.

IN USE characterize the thin film

CVD

Chemical vapor deposition, forming a film through chemical reactions of vapor precursors.

IN USE evaluate CVD film properties

PVD

Physical vapor deposition, transferring material through a physical vapor process.

IN USE review PVD performance

ALD

Atomic layer deposition, using successive self-limiting surface reactions.

IN USE assess ALD conformality

precursor

A starting substance used in a film-forming reaction.

IN USE qualify the precursor

conformality

How consistently a film coats different surface orientations or features.

IN USE assess film conformality

step coverage

Relative film coverage over a feature's steps or sidewalls.

IN USE evaluate step coverage

film stress

Internal stress within a deposited film.

IN USE measure film stress

thickness nonuniformity

Variation in film thickness under a defined metric.

IN USE report thickness nonuniformity

etch selectivity

Relative removal rates of different materials in an etch process.

IN USE assess etch selectivity

anisotropy

Direction-dependent behavior, such as preferential vertical removal.

IN USE evaluate etch anisotropy

Precision check

The metric improves by 2 percentage points, a 40% relative reduction from its initial 5% value. Neither calculation establishes the proposed window. The same measurement definition is used here; other nonuniformity formulas must not be mixed into the comparison.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

endpoint detection

Monitoring used to identify a process completion condition.

IN USE verify endpoint detection

overetch

Additional etching beyond a defined clearing or endpoint condition.

IN USE assess overetch effects

CMP

Chemical mechanical planarization, combining chemical and mechanical surface removal.

IN USE evaluate CMP compatibility

slurry

A fluid mixture used in polishing or planarization.

IN USE qualify the CMP slurry

dishing

Local depression of a material surface during planarization.

IN USE measure dishing

erosion

Excess regional material removal during planarization.

IN USE assess erosion

process window

A range of conditions meeting defined process requirements.

IN USE establish the process window

nominal setting

The chosen central or reference process setting.

IN USE test the nominal setting

corner condition

A selected limiting combination in an evaluation range.

IN USE evaluate corner conditions

process interaction

An effect arising from the relationship between process steps or factors.

IN USE investigate process interactions

integration trade-off

An exchange between benefits and effects across process steps.

IN USE explain the integration trade-off

recipe release

Authorization of a controlled process recipe for defined use.

IN USE obtain recipe release

Precision check

A deposition improvement can affect later etch and planarization behavior. The case supplies no result for those steps. Describe their assessment as pending, not as either a guaranteed benefit or a demonstrated new failure.

Phrases you can actually use

C Notice the language / 1 of 2

03

IDENTIFY THE METRIC

"The defined thickness-nonuniformity metric fell from 5% to 3%."

STATE THE CONDITION

"The result applies to the nominal setting only."

LIMIT THE CLAIM

"One center point does not establish a process window."

CHECK INTERACTIONS

"What does the film change mean for the subsequent etch?"

KEEP THE BASIS FIXED

"Both values use the same measurement definition."

SEPARATE CHANGES

"That is two percentage points, or a 40% relative reduction."

NAME THE MISSING TESTS

"The proposed low and high settings are untested."

ASK ABOUT PLANARIZATION

"Has compatibility with the CMP step been assessed?"

Uniformity

State the formula or metric, not only a percentage.

Window

A range meeting requirements, not one successful setting.

Corner

A defined edge or combination in the evaluation plan.

Phrases you can actually use

C Notice the language / 2 of 2

03

AVOID A CAUSAL LEAP

"Better uniformity does not prove every film property improved."

QUALIFY MATURITY

"This is a trial result, not a released recipe."

PRESERVE CONFIGURATION

"Keep the trial recipe revision with the measurement record."

AVOID DECLARING FAILURE

"Untested corners are unresolved, not failed."

IDENTIFY TRADE-OFFS

"We still need the integration trade-off assessment."

ASK FOR CRITERIA

"Which responses must meet limits across the range?"

REQUEST BOUNDED FOLLOW-UP

"Assess the planned range and the identified downstream effects."

CLOSE THE SUMMARY

"Report the center-point result and the remaining validation scope."

Better

Name the response that improved and what was not assessed.

Integrated

Requires consideration of connected process steps.

Released

An authorization status, not a synonym for promising.

Choose the precise message

D Check your understanding

03

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which summary preserves the evidence boundary?

- A** Production release follows automatically from the lower metric.
- B** The nominal-setting metric improved; the range and downstream effects remain untested.
- C** The full process window is validated by the center result.
- D** All etch and CMP outcomes improved by 40%.

2. What is the relative reduction from 5% to 3%?

- A** 60%
- B** 3%
- C** 40%
- D** 2%

3. Why review downstream etch and CMP?

- A** Their results can be assumed from the recipe name.
- B** A film change may alter later process behavior.
- C** Deposition results never interact with later operations.
- D** Lower nonuniformity proves all later steps are equivalent.

4. Which status should remain separate from trial success?

- A** The recorded original value of 5%
- B** Recipe release
- C** The reported nominal-setting value of 3%
- D** The unchanged metric definition

Answers and explanations: [page 80](#).

A better center point is not a wider window

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

integration trade-off / CMP / recipe release / thickness nonuniformity / process window / process interaction / deposition / etch selectivity / corner condition / nominal setting

- 01 **Yuki:** The nominal-setting trial reduced our thickness-nonuniformity metric from five percent to three percent. The summary now says the full process window is validated, which seems too broad.
- 02 **Sam:** It is too broad. A ¹ _____ covers a range of conditions meeting defined requirements. We have one encouraging center-point result, not evidence for the entire proposed range.
- 03 **Yuki:** Both values use the same measurement definition, and lower is better for this metric. I want to report that improvement clearly without losing the limitation.
- 04 **Sam:** Name ² _____ explicitly. Saying the process improved could imply that every relevant response was assessed, while the supplied result addresses only this defined metric.
- 05 **Yuki:** The absolute decrease is two percentage points. Relative to the original five percent, the decrease is forty percent, not merely two percent.
- 06 **Sam:** Keep that arithmetic tied to the ³ _____. Neither way of expressing the improvement extends the evidence to settings that have not been run.
- 07 **Yuki:** The proposed low and high settings are still untested. We need to avoid describing them as passed, but their absence also does not prove they would fail.
- 08 **Sam:** Correct. Each planned ⁴ _____ needs its own assessment under the approved plan. An untested condition is unresolved, not a favorable result or a demonstrated defect.
- 09 **Yuki:** The deposition change may also affect other film characteristics. A lower thickness-variation metric does not tell us that stress, coverage, or later pattern transfer remained acceptable.
- 10 **Sam:** That is an ⁵ _____ question. We need to assess the relevant responses and interactions rather than optimize one number in isolation from the product flow.

The conversation, continued

E Conversations / turns 11-20

03

Continue with the word bank on page 28.

- 11 **Yuki:** For the subsequent etch, the team has not yet checked how the changed film behaves relative to the neighboring materials. That could affect the interpretation of the trial.
- 12 **Sam:** Review ⁶ _____ where it matters to the approved integration assessment. Naming the concern identifies a question to investigate, not a result we can claim already occurred.
- 13 **Yuki:** The planarization team also needs the proposed film information. Their review has not started, so I should not imply that the next surface-finishing step is already covered.
- 14 **Sam:** State that ⁷ _____ compatibility remains open. The deposition result does not establish that planarization behavior, surface effects, or related requirements are unchanged.
- 15 **Yuki:** We can identify the relevant responses in the evaluation plan without improvising operating settings during this review. The process owners should define the approved technical work.
- 16 **Sam:** Yes. Capture each ⁸ _____ question and its owner. That makes the remaining work specific enough to schedule and review without treating the meeting as recipe authorization.
- 17 **Yuki:** The recipe is still a trial revision. Production teams should not receive the summary as though the new settings were ready for routine use.
- 18 **Sam:** Keep ⁹ _____ separate from the experiment. A favorable metric can support the review, but the required scope and approvals still need to be completed.
- 19 **Yuki:** I will revise the summary to report the center-point improvement, the unchanged metric basis, and the untested range. I will list the etch and planarization reviews as pending.
- 20 **Sam:** That presents the ¹⁰ _____ result accurately. The team can recognize progress while seeing exactly what evidence is still needed before making a broader process claim.

ANSWERS / EXPLANATIONS: PAGE 81

1. process window | 2. thickness nonuniformity | 3. nominal setting | 4. corner condition | 5. integration trade-off | 6. etch selectivity | 7. CMP | 8. process interaction | 9. recipe release | 10. deposition

An improved metric with a limited range

F Say it

At one nominal setting, an unchanged nonuniformity metric falls from 8% to 6%. Lower is better. Edge settings and downstream etch remain untested; the recipe is not released.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

25 / release / 2 / nominal

Engineer: "The decrease is ¹ _____ percentage points."

Reviewer: "The relative reduction is ² _____ percent."

Engineer: "Only the ³ _____ setting was tested."

Reviewer: "The recipe still needs ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 81

1. 2 | 2. 25 | 3. nominal | 4. release

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Two percentage points, not a proven process gain

A Read the situation

04

Report yield with counts and comparison limits, and distinguish product acceptance from evidence about process stability.

Yield engineer Ravi presents a pilot lot with 92 passing die out of 100 tested, compared with a reference lot with 900 passing die out of 1,000 tested. The product mixes differ, and no controlled comparison or statistical significance assessment has been completed. Manager Elena wants to announce that a process change caused a two-percent yield improvement. The team also has no time-ordered control-chart analysis supporting a stability claim. Ravi must preserve the useful observations while correcting the causal, percentage, and stability language.

PEOPLE / Elena: Yield program manager | Ravi: Yield engineer

GIVE THE COUNTS BEFORE THE CONCLUSION

A favorable yield difference can start an investigation without ending it. State the passing and tested counts, identify changed product mix, and distinguish an absolute percentage-point difference from a relative percentage change. Avoid using the words significant or stable as casual substitutes for better.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is the pilot yield?

- A 100% because the pilot completed
- C 90% from 900 of 1,000 die

- B 92% from 92 of 100 tested die
- D 2% from two lots

2. What is the absolute difference between the observed yields?

- A Exactly 2 relative percent
- C 2 percentage points

- B 20 percentage points
- D 2 passing die

3. Which factor limits the causal comparison?

- A The pilot is called a pilot.
- C The product mixes differ.

- B The reference contains more passing die in absolute number.
- D Both yields have denominators.

Answers and explanations: page 82.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

metrology

The science and practice of measurement.

IN USE review metrology consistency

electrical yield

The proportion meeting defined electrical test criteria.

IN USE report electrical yield

pass count

The number of tested items meeting the stated criteria.

IN USE preserve the pass count

tested population

The set of items included in a reported test result.

IN USE define the tested population

denominator

The total used as the base of a fraction or percentage.

IN USE state the denominator

product mix

The composition of different products in a dataset.

IN USE adjust for product mix

pilot lot

A limited production lot used to evaluate a process or product.

IN USE evaluate a pilot lot

reference lot

A lot used as a stated comparison basis.

IN USE identify the reference lot

percentage point

A unit for the absolute difference between percentages.

IN USE report percentage-point change

relative change

A change divided by its specified starting value.

IN USE calculate relative change

confounding factor

A factor that prevents clean attribution of an observed relationship.

IN USE identify a confounding factor

stratification

Separating data into relevant comparable groups.

IN USE stratify by product

Precision check

The pilot has 92% yield and the reference 90%. The difference is 2 percentage points, or about 2.22% relative to 90%. Different product mixes and sample sizes mean this observation alone does not establish a causal improvement or statistical significance.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

SPC

Statistical process control, using data to monitor process behavior.

IN USE apply SPC

control chart

A time-ordered chart with a defined process-monitoring method.

IN USE review the control chart

control limit

A monitoring boundary derived under a stated process model or method.

IN USE distinguish control limits

specification limit

A defined product or process acceptance boundary.

IN USE check specification limits

common-cause variation

Variation inherent in the currently defined process system.

IN USE characterize common-cause variation

special-cause variation

Variation associated with a nonroutine identifiable influence.

IN USE investigate special-cause variation

control signal

A result meeting a defined rule for investigation on a control chart.

IN USE respond to a control signal

process stability

Consistent statistical behavior under the defined monitoring basis.

IN USE assess process stability

subgroup

A defined set of observations grouped for process analysis.

IN USE define rational subgroups

measurement system

The instruments, methods, people, and conditions producing measurements.

IN USE assess the measurement system

yield learning

Investigation that improves understanding of yield losses and drivers.

IN USE support yield learning

causal attribution

Assigning an observed effect to a particular cause.

IN USE qualify causal attribution

Precision check

Control limits describe expected process behavior under a defined monitoring method; specification limits express acceptance requirements. Passing a specification does not establish stability, and a control signal does not automatically identify either the cause or the disposition of every unit.

Phrases you can actually use

C Notice the language / 1 of 2

GIVE THE PILOT COUNTS

"The pilot passed 92 of 100 tested die."

NAME THE ABSOLUTE CHANGE

"The observed difference is two percentage points."

EXPOSE THE MIX

"The two lots have different product mixes."

QUALIFY SIGNIFICANCE

"Statistical significance has not been assessed."

GIVE THE REFERENCE COUNTS

"The reference passed 900 of 1,000 tested die."

NAME THE RELATIVE BASIS

"Relative to 90%, that is approximately 2.22%."

AVOID CAUSALITY BY TIMING

"The process change is not isolated by this comparison."

REQUEST LIKE-FOR-LIKE DATA

"Compare relevant product groups on a consistent basis."

Improvement

Can describe an observation or an established effect; specify which.

Significant

Distinguish business importance from statistical evidence.

Stable

Needs a defined time-ordered process assessment.

Phrases you can actually use

C Notice the language / 2 of 2

SEPARATE LIMITS

"Control limits and specification limits serve different purposes."

KEEP THE METHOD VISIBLE

"State the subgrouping and monitoring rules."

AVOID INVENTED DIAGNOSIS

"The signal does not identify its own cause."

CHECK MEASUREMENT CONSISTENCY

"Confirm the measurement and test basis before combining data."

ASK FOR TIME ORDER

"Which time-ordered evidence supports the stability claim?"

AVOID DISMISSING A SIGNAL

"A control signal warrants investigation under the defined procedure."

PRESERVE USEFUL PROGRESS

"The pilot result is encouraging but limited."

CLOSE THE FINDING

"Report the counts, observed difference, limitations, and next analysis."

Yield

Attach the test stage, criteria, and denominator.

In spec

An acceptance statement, not proof of statistical control.

Caused

Stronger than followed or coincided with.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which announcement is supported?

- A** The process change is proven to improve every product by two relative percent.
- B** The pilot proves the process is statistically stable.
- C** Different denominators make both observed percentages meaningless.
- D** The pilot observed 92% versus 90% yield, with different product mixes and no causal conclusion.

2. What is the approximate relative increase from 90% to 92%?

- A** 92.00%
- B** 2.22%
- C** 2.00%
- D** 20.00%

3. Which next analysis addresses the product-mix problem?

- A** Assume all products have identical yield without checking.
- B** Separate comparable product groups and review consistent test bases.
- C** Remove the reference denominator from the slide.
- D** Rename the pilot a production lot.

4. Which statement distinguishes control and specification limits?

- A** Control limits monitor behavior; specification limits define acceptance boundaries.
- B** They are always identical and interchangeable.
- C** A specification pass proves no special cause exists.
- D** A control signal alone proves all material must be scrapped.

Answers and explanations: [page 82](#).

Two percentage points, not a proven process gain

E Conversations / turns 1-10

04

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

percentage point / control signal / specification limit / yield learning / stratification / product mix / measurement system / causal attribution / process stability / pass count

- 01 **Elena:** The pilot reached ninety-two percent against ninety percent in the reference lot. Can we announce that the process change caused a two-percent yield improvement?
- 02 **Ravi:** Begin with the ¹ _____ and denominator. The pilot passed ninety-two of one hundred tested die; the reference passed nine hundred of one thousand.
- 03 **Elena:** Those counts make the sample-size difference visible. They do not stop us from reporting the observed yields, but they matter when we interpret the comparison.
- 04 **Ravi:** Correct. Also state the ² _____. The lots contain different product compositions, so the two percentages are not a controlled comparison of the process change alone.
- 05 **Elena:** The wording about two percent needs correction too. The absolute difference between ninety and ninety-two percent is two percentage points.
- 06 **Ravi:** Yes, use ³ _____ for that absolute change. Relative to the ninety-percent starting value, the increase is approximately two point two two percent.
- 07 **Elena:** I want the team to see the pilot as useful progress without overstating what caused the result. The change happened before the pilot, but timing is not enough.
- 08 **Ravi:** Avoid unsupported ⁴ _____. The result can guide further investigation while the product mix and other relevant differences remain visible.
- 09 **Elena:** We should compare the relevant products separately and confirm their test basis. Otherwise a favorable shift in composition might look like a process benefit.
- 10 **Ravi:** Use ⁵ _____ to examine the appropriate groups. It will not automatically prove causality, but it addresses one important weakness in the aggregate comparison.

The conversation, continued

E Conversations / turns 11-20

04

Continue with the word bank on page 37.

- 11 **Elena:** The draft also says the pilot demonstrates stable production. We have not presented a time-ordered analysis, only these two lot summaries.
- 12 **Ravi:** Then remove that ⁶ _____ claim. A favorable summary percentage does not establish how the process behaves over time or whether a monitoring rule has signaled a change.
- 13 **Elena:** We have acceptance criteria for individual test results. Some colleagues are treating those limits as though they were the same as the chart's monitoring boundaries.
- 14 **Ravi:** A ⁷ _____ defines an acceptance boundary. A control limit has a different purpose, so passing the specification is not proof that the process is in statistical control.
- 15 **Elena:** If a monitoring rule signals something unusual, the team needs to investigate. But the signal itself does not identify which tool or material caused it.
- 16 **Ravi:** Exactly. A ⁸ _____ is a prompt for the defined investigation process. Do not convert it directly into a root-cause statement or an automatic disposition for every unit.
- 17 **Elena:** We also need to make sure the measurements and test criteria are consistent before combining records. Otherwise the next comparison could carry a different hidden change.
- 18 **Ravi:** Check the ⁹ _____ and test basis. Clear denominators are essential, but they do not solve inconsistencies in how observations were produced or classified.
- 19 **Elena:** The revised update will report the counts, the two-point difference, and the changed product mix. It will leave significance, stability, and causality unclaimed pending the appropriate analysis.
- 20 **Ravi:** That supports ¹⁰ _____. The pilot remains informative, and the team gets a precise next question instead of a success claim that the present data cannot establish.

ANSWERS / EXPLANATIONS: PAGE 83

1. pass count | 2. product mix | 3. percentage point | 4. causal attribution | 5. stratification | 6. process stability | 7. specification limit | 8. control signal | 9. measurement system | 10. yield learning

Percentage points need a denominator

F Say it

A trial passes 48 of 50 tested die, or 96%. A reference passes 450 of 500, or 90%. Product mixes differ, and no causal analysis is complete.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

50 / unproven / 500 / 6

Engineer: "The trial denominator is ¹ _____."

Reviewer: "The reference denominator is ² _____."

Engineer: "The absolute yield difference is ³ _____ percentage points."

Reviewer: "A causal process improvement remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 83

1. 50 | 2. 500 | 3. 6 | 4. unproven

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Twenty-four detections are not a confirmed source

A Read the situation

Describe inspection findings with location, area, timing, and classification while avoiding premature contamination-source claims.

Inspection engineer Hana and contamination-control engineer Oscar review a note saying cleanroom problem. The underlying record contains 24 particle-like detections over 200 square centimeters inspected on wafers from lot P8 after tool C2 between 9 and 10 a.m. The map shows an edge cluster. No comparable pre-process scan exists, defect review is incomplete, and no source is confirmed. The lot is held under the local procedure. The team must distinguish detections, confirmed defects, added defects, and airborne particle measurements.

PEOPLE / Hana: Inspection engineer | Oscar: Contamination-control engineer

DESCRIBE THE PATTERN BEFORE NAMING THE SOURCE

A location pattern can focus an investigation without proving where contamination originated. State the observation and its limits, then request traceable process and inspection records. Avoid turning a vague cleanroom label into blame for a team, operator, or tool before the evidence supports it.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What does the inspection record contain?

A 24 particle-like detections over 200 square centimeters

C An airborne count per cubic meter

B 24 confirmed electrical failures on every wafer

D A measured increase of 24 defects from a matched pre-scan

2. What does the map show?

A Uniform distribution across every layer

C A completed source elimination study

B A confirmed cleanroom air source

D An edge cluster

3. Why cannot the 24 detections be called 24 added defects?

A The lot has an identifier.

C No comparable pre-process scan exists, and review is incomplete.

B Surface maps always measure airborne particles.

D The inspected area is known.

Answers and explanations: page 84.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

inspection detection

A feature flagged by an inspection method for assessment.

IN USE review inspection detections

defect review

Examination used to classify and interpret detected features.

IN USE complete defect review

defect density

Defects per unit area on a stated counting and inspection basis.

IN USE report defect density

inspected area

The surface area included in an inspection result.

IN USE state the inspected area

particle-like signature

An observed appearance consistent with a particle, pending review.

IN USE classify a particle-like signature

nuisance detection

A flagged feature not relevant to the defect concern being assessed.

IN USE filter nuisance detections

killer defect

A defect that causes functional failure in its particular context.

IN USE assess killer-defect relevance

defect classification

Assignment of detected features to defined categories.

IN USE refine defect classification

wafer map

A spatial representation of results on a wafer.

IN USE examine the wafer map

edge cluster

A concentration of observations near a wafer's edge.

IN USE investigate an edge cluster

spatial signature

A location pattern in measured or inspected results.

IN USE compare spatial signatures

added defect

A defect introduced between comparable inspection points.

IN USE quantify added defects

Precision check

The observed detection density is 24 divided by 200, or 0.12 detections per square centimeter. Do not call it a confirmed killer-defect density. Classification is incomplete, and no comparable pre-scan establishes how many detections were introduced by C2.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

pre-process scan

Inspection performed before a specified process step.

IN USE preserve the pre-process scan

post-process scan

Inspection performed after a specified process step.

IN USE compare post-process scans

airborne particle count

A count of particles in a defined air sample or volume.

IN USE review airborne particle counts

surface contamination

Unwanted material present on a surface.

IN USE investigate surface contamination

cross-contamination

Transfer of unwanted material between areas, items, or processes.

IN USE prevent cross-contamination

contamination source

The origin of unwanted material.

IN USE identify the contamination source

particle excursion

An unusual particle-related result under defined monitoring rules.

IN USE investigate a particle excursion

inspection sensitivity

The ability of an inspection method to detect specified features.

IN USE confirm inspection sensitivity

size threshold

The size boundary used for counting or classifying features.

IN USE state the size threshold

exposure history

The record of relevant environments and handling encountered.

IN USE reconstruct exposure history

suspect interval

A defined period being examined for possible affected material.

IN USE bound the suspect interval

containment scope

The material or operations covered by a temporary restriction.

IN USE define containment scope

Precision check

A surface result per square centimeter is not an airborne count per cubic meter. Measurement type, inspected area or air volume, size threshold, and sensitivity belong with the number. Similar timing or spatial patterns guide investigation but do not prove a source.

Phrases you can actually use

C Notice the language / 1 of 2

REPLACE THE VAGUE LABEL

"The record shows particle-like surface detections, not a confirmed cleanroom source."

NAME THE DENSITY BASIS

"That is 0.12 detections per square centimeter."

BOUND THE INTERVAL

"These records follow C2 between 9 and 10 a.m."

AVOID AN ADDED-DEFECT CLAIM

"We lack a comparable pre-process scan."

GIVE THE COUNT AND AREA

"There were 24 detections over 200 square centimeters inspected."

LOCATE THE PATTERN

"The wafer map shows an edge cluster."

PRESERVE THE UNCERTAINTY

"Defect classification remains incomplete."

SEPARATE MEASUREMENT TYPES

"Surface detections are not airborne particle counts."

Particle-like

An appearance classification, not necessarily a confirmed material identity.

Added

Needs comparable before-and-after evidence.

Density

Requires a count definition and area or volume denominator.

Phrases you can actually use

C Notice the language / 2 of 2

ASK ABOUT SENSITIVITY

"Were the inspection settings and size threshold consistent?"

TRACE THE MATERIAL

"Link the map to the lot, wafers, and process history."

DEFINE THE NEXT REVIEW

"Review the detections and relevant exposure history."

ASK ABOUT SCOPE

"Which material is included in the current containment?"

AVOID PREMATURE BLAME

"The spatial signature does not confirm the source."

KEEP THE RESTRICTION CLEAR

"P8 remains held under the local procedure."

AVOID AUTOMATIC YIELD CLAIMS

"Not every flagged feature is a confirmed killer defect."

CLOSE THE NOTE

"Report observations, limits, containment status, and investigation ownership."

Source

A causal conclusion, not merely the last tool visited.

Cluster

A spatial pattern that can guide but not finish an investigation.

Cleanroom problem

Too broad to locate the evidence or assign a justified cause.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. What is the observed detection density?

- A** 0.12 particles per cubic meter of air
- B** 0.12 detections per square centimeter
- C** 12 detections per square centimeter
- D** 8.33 detections per square centimeter

2. Which conclusion is unsupported?

- A** The records are after C2.
- B** The map shows an edge cluster.
- C** Defect review is incomplete.
- D** C2 introduced all 24 detections.

3. Which next record is useful for the investigation?

- A** A density with its denominator removed
- B** A new source conclusion before reviewing detections
- C** Traceable wafer, inspection-setting, timing, and exposure information
- D** An operator's nationality as an assumed contamination cause

4. Which distinction is correct?

- A** Surface detection density and airborne particle counts use different measurement bases.
- B** A square centimeter and a cubic meter are interchangeable denominators.
- C** Every surface detection is a proven electrical failure.
- D** An edge cluster proves an airborne source.

Answers and explanations: [page 84](#).

Twenty-four detections are not a confirmed source

E Conversations / turns 1-10

05

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

exposure history / airborne particle count / edge cluster / contamination source / Defect review / containment scope / inspected area / pre-process scan / added defect / inspection sensitivity

- 01 **Hana:** The review note says cleanroom problem, but that label is much broader than the evidence. We have twenty-four particle-like detections on the inspected surfaces from P8.
- 02 **Oscar:** Start with the ¹ _____. The record covers two hundred square centimeters, so the number can be interpreted on a defined surface basis rather than as an unexplained total.
- 03 **Hana:** That gives zero point one two detections per square centimeter. I should keep the word detections because the detailed classification has not been completed.
- 04 **Oscar:** Correct. ² _____ is still open. A flagged feature is not automatically a confirmed defect of the relevant type, much less a demonstrated electrical failure.
- 05 **Hana:** The wafer map shows the observations concentrated near the edge. That pattern may help the investigation, but it does not identify a source by itself.
- 06 **Oscar:** Call it an ³ _____. Describe where it appears and connect the map to the wafer IDs, rather than turn the pattern into an immediate diagnosis of the cleanroom.
- 07 **Hana:** The records follow C2 between nine and ten a.m. Someone has rewritten that as C2 added twenty-four particles, although we do not have a comparable earlier scan.
- 08 **Oscar:** We cannot claim an ⁴ _____ count on that basis. After C2 establishes the recorded sequence, not proof that C2 introduced every detected feature.
- 09 **Hana:** A matched earlier inspection would help distinguish what was already present from what appeared later. We would also need to check the inspection basis before comparing counts.
- 10 **Oscar:** Exactly. A ⁵ _____ must be comparable with the later scan. Different sensitivity, locations, or thresholds could otherwise make a count difference look like process-added contamination.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 46.

05

- 11 **Hana:** The facilities team has separate air-monitoring information. We should not place those values in the same column as the surface counts without naming their units and methods.
- 12 **Oscar:** An ⁶ _____ is a different measurement. A value per air volume cannot be treated as the same quantity as detections per inspected surface area.
- 13 **Hana:** I will include the inspection recipe and threshold with the maps. That will let reviewers see whether the method was capable of detecting the features being discussed.
- 14 **Oscar:** Keep ⁷ _____ visible. A tool finding more features after a settings change does not by itself prove that the process produced more contamination.
- 15 **Hana:** P8 remains on hold under the local procedure. The investigation team needs to say which material and time interval are included, rather than leave the restriction implicit.
- 16 **Oscar:** Define the ⁸ _____ through the authorized process. A precise scope supports coordination without turning this discussion into permission to release or extend the hold arbitrarily.
- 17 **Hana:** We can reconstruct the lot's recent handling and process history alongside the spatial pattern. That should help distinguish possible explanations and identify useful follow-up evidence.
- 18 **Oscar:** Review the ⁹ _____. It can guide source investigation, but a shared tool or time interval is not enough on its own to establish the origin.
- 19 **Hana:** The corrected note will give the count, area, edge pattern, interval, and incomplete classification. It will state that introduction by C2 and an airborne source are unconfirmed.
- 20 **Oscar:** That keeps the ¹⁰ _____ question open on an accurate basis. We can investigate efficiently while preserving the distinction between an observation, a hypothesis, and a confirmed cause.

ANSWERS / EXPLANATIONS: PAGE 85

1. inspected area | 2. Defect review | 3. edge cluster | 4. added defect | 5. pre-process scan | 6. airborne particle count | 7. inspection sensitivity | 8. containment scope | 9. exposure history | 10. contamination source

Count the detections, not an invented cause

F Say it

A post-process scan flags 12 features over 100 square centimeters. Classification is pending, and no comparable pre-process scan exists. No contamination source is confirmed.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

0.12 / pre-process / pending / unconfirmed

Engineer: "The observed density is ¹ _____ detections per square centimeter."

Reviewer: "Feature classification is still ² _____."

Engineer: "There is no comparable ³ _____ scan."

Reviewer: "The contamination source is ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 85

1. 0.12 | 2. pending | 3. pre-process | 4. unconfirmed

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Same recipe, different means

A Read the situation

Distinguish recipe identity, specification performance, tool matching, and equipment availability in a transfer discussion.

Equipment engineer Victor and process engineer Asha compare tools A and B using the same identified R7 recipe and an agreed comparable measurement basis. Their film-thickness sample means are 98 nm and 102 nm. The local mean specification is 95 to 105 nm, but the tool-matching criterion requires the means to differ by no more than 2 nm. Tool B was available for 900 of a defined 1,000-minute window. A colleague uses the identical recipe and 90% availability to claim B is an approved interchangeable backup. No transfer approval is recorded.

PEOPLE / Victor: Equipment engineer | Asha: Process engineer

06

NAME WHICH READINESS QUESTION YOU ANSWERED

Available, in specification, matched, and released describe different conditions. A tool may satisfy one and not another. Give each claim its own evidence and criterion so production planning does not convert equipment uptime into process-transfer approval.

Briefing checks

Circle one answer in each item. Use the briefing.

1. How far apart are the sample means?

A 2 nm

B 7 nm

C 200 nm

D 4 nm

2. Do the means meet the stated matching criterion?

A Yes; both tools use R7.

B The availability percentage automatically settles matching.

C No; 4 nm exceeds the allowed 2 nm difference.

D Yes; both means are within the mean specification.

3. What does 90% describe here?

A Availability for 900 of the defined 1,000 minutes

B The proportion of wafers passing final test

C A 90% probability that the tools are interchangeable

D A recorded production transfer approval

Answers and explanations: page 86.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

tool matching

Demonstrating comparable tool performance against defined criteria.

IN USE assess tool matching

chamber matching

Demonstrating comparable performance between process chambers.

IN USE verify chamber matching

recipe revision

The identified version of a process recipe.

IN USE confirm the recipe revision

setpoint

A commanded target value for a controlled variable.

IN USE record the setpoint

actual value

The measured or achieved value rather than the commanded target.

IN USE compare setpoint and actual value

equipment state

The recorded operational condition of a tool.

IN USE report equipment state

availability

The share of a defined time basis during which equipment is available.

IN USE calculate equipment availability

utilization

The share of a stated time basis during which equipment is used.

IN USE state the utilization denominator

scheduled downtime

Planned time when equipment is unavailable for the defined use.

IN USE plan scheduled downtime

unscheduled downtime

Unplanned time when equipment is unavailable.

IN USE investigate unscheduled downtime

preventive maintenance

Planned work intended to reduce equipment failures or deterioration.

IN USE schedule preventive maintenance

corrective maintenance

Work to restore equipment after a fault or failure.

IN USE complete corrective maintenance

Precision check

Both sample means meet the local 95-to-105 nm mean specification. Their difference is 4 nm, so they fail the separate 2 nm matching criterion. This does not establish the status of every individual wafer, whose measurements are not supplied.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

MTBF

Mean time between failures under a stated operating and event definition.

IN USE report MTBF

MTTR

Mean time to repair or restore under the stated local definition.

IN USE define the MTTR basis

chamber condition

The physical and operating state of a process chamber.

IN USE assess chamber condition

seasoning

Controlled processing used to establish a required chamber condition.

IN USE verify seasoning status

calibration status

The current documented state of required measurement calibration.

IN USE check calibration status

qualification wafer

A wafer used to assess a specified tool or process condition.

IN USE evaluate a qualification wafer

monitor wafer

A wafer used to track selected process or tool behavior.

IN USE review monitor-wafer results

matching criterion

A defined limit for acceptable performance difference.

IN USE apply the matching criterion

process transfer

Moving a defined process to another tool or location.

IN USE approve process transfer

backup tool

An alternative tool intended for use under defined conditions.

IN USE qualify a backup tool

release status

Whether the required authorization for a specified use is in place.

IN USE confirm release status

offset correction

A controlled adjustment addressing a measured systematic difference.

IN USE assess offset correction

Precision check

Availability is 900 divided by 1,000, or 90%, on this stated basis. It does not measure process matching, output yield, or production-use approval. Availability and utilization definitions vary, so keep the time denominator explicit.

Phrases you can actually use

C Notice the language / 1 of 2

CONFIRM THE RECIPE

"Both tools used the identified R7 revision."

STATE THE MEANS

"A measured 98 nm and B measured 102 nm."

APPLY THE RIGHT CRITERION

"Matching allows no more than a 2 nm difference."

BOUND THE EVIDENCE

"These are sample means, not every individual wafer result."

SEPARATE INPUTS FROM OUTPUTS

"Identical setpoints do not prove identical performance."

GIVE THE DIFFERENCE

"The means are 4 nm apart."

SEPARATE THE SPECIFICATIONS

"Both means are in spec, but the pair is not matched."

ASK ABOUT CHAMBER STATE

"Were the relevant chamber conditions documented?"

Same

Identify whether recipe, settings, product, or measured outcome is the same.

Available

Describes a time or operational state, not qualification.

In spec

Specify the metric; a passing mean does not prove every unit passed.

Phrases you can actually use

C Notice the language / 2 of 2

CHECK MEASUREMENT STATUS

"Confirm the calibration and measurement basis."

KEEP THE PERCENTAGE LIMITED

"That 90% figure is availability, not yield."

ROUTE THE INVESTIGATION

"Review the observed offset before proposing a correction."

CLARIFY THE TRANSFER

"Which process, product, and tool combination is proposed?"

STATE AVAILABILITY

"B was available for 900 of the defined 1,000 minutes."

AVOID AUTOMATIC SUBSTITUTION

"Backup status needs the required qualification and approval."

PRESERVE AUTHORIZATION

"Do not change the recipe from this comparison alone."

CLOSE THE STATUS

"Report specification, matching, availability, and release separately."

06

Matched

Requires an actual matching criterion and supporting evidence.

Backup

A planning role that does not automatically authorize production use.

Offset

A measured difference to investigate, not an instruction to adjust settings.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

06

1. Which status report is accurate?

- A** Tool B is approved because its availability is 90%.
- B** Every wafer from both tools is proven in specification.
- C** Both means meet the mean specification; their 4 nm difference fails the 2 nm matching limit.
- D** Both tools are matched because 98 and 102 lie within 95 to 105.

2. What does using the same recipe establish?

- A** All tool-to-tool differences are zero.
- B** The identified recipe input is shared, not that output performance is interchangeable.
- C** The chamber conditions must be identical.
- D** Every calibration requirement has been met.

3. Which calculation matches the availability basis?

- A** 102 minus 98 equals 90%.
- B** The film-thickness specification directly determines uptime.
- C** 900 divided by 1,000 equals 90%.
- D** 1,000 divided by 900 equals 90%.

4. What remains necessary before calling B an approved interchangeable backup?

- A** Only a passing sample mean against the broad mean specification
- B** The required matching or qualification evidence and transfer approval
- C** Only a repeated statement that R7 is installed
- D** Only a higher availability percentage

Answers and explanations: [page 86](#).

Same recipe, different means

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

matching criterion / Tool matching / release status / chamber condition / calibration status / availability / equipment state / process transfer / actual value / recipe revision

06

- 01 **Victor:** Tool B has the same R7 recipe as A and was available for ninety percent of the window. The planning note calls it an interchangeable backup.
- 02 **Asha:** That combines different questions. ¹ _____ depends on the measured outcomes and our criterion, not only the recipe name or the amount of time the tool was available.
- 03 **Victor:** We confirmed the same recipe revision and an agreed measurement basis. A's sample mean was ninety-eight nanometers, while B's was one hundred two.
- 04 **Asha:** The ² _____ allows a difference of no more than two nanometers. These means differ by four, so this comparison does not establish an acceptable match.
- 05 **Victor:** Both means are within the local mean specification of ninety-five to one hundred five. That explains why someone thought the comparison had passed.
- 06 **Asha:** The ³ _____ has to be assessed against the correct question. Each mean passes that specification, but the pair fails the separate matching limit.
- 07 **Victor:** We should also avoid saying every wafer was in specification. The figures we have in this review are sample means, not the complete individual measurements.
- 08 **Asha:** Agreed. Confirm the ⁴ _____ and report the evidence at its actual level. Do not let a summary mean or a shared input become a claim about every processed unit.
- 09 **Victor:** The observed difference needs investigation. Chamber history and maintenance condition may be relevant, but neither has been established as the cause from these numbers.
- 10 **Asha:** Review ⁵ _____ and the other appropriate records. Identical commanded settings do not guarantee identical physical conditions or measurement behavior.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 55.

- 11 **Victor:** Calibration and monitor records should be linked to the comparison as well. That will help the process team assess whether the apparent offset has a consistent basis.
- 12 **Asha:** Check ⁶ _____ through the approved process. We should not prescribe a recipe adjustment before the team understands the relevant evidence and authorization requirements.
- 13 **Victor:** For the uptime figure, B was available for nine hundred minutes out of a defined one-thousand-minute window. That calculation gives ninety percent.
- 14 **Asha:** Label it ⁷ _____ and retain the denominator. It does not mean ninety percent yield or ninety percent confidence that the tool can substitute for A.
- 15 **Victor:** We also have no usage-time figure here, so we should not relabel availability as utilization. The two measures can use different time bases and answer different questions.
- 16 **Asha:** Correct. The ⁸ _____ record helps explain the time measure, but an available tool is not automatically qualified for this product and process combination.
- 17 **Victor:** The backup proposal remains useful because capacity is constrained, but it needs the appropriate technical assessment and approval before production relies on it.
- 18 **Asha:** Treat it as a ⁹ _____ decision. The defined scope, matching evidence, and required authorization must be clear instead of being inferred from the existence of an alternative tool.
- 19 **Victor:** I will revise the planning note to separate the passing means, failed matching comparison, ninety-percent availability, and missing approval. It will no longer call B interchangeable.
- 20 **Asha:** That preserves the ¹⁰ _____ accurately. We can pursue the backup option while making its unresolved requirements visible to the people planning production.

06

ANSWERS / EXPLANATIONS: PAGE 87

1. Tool matching | 2. matching criterion | 3. actual value | 4. recipe revision | 5. chamber condition | 6. calibration status | 7. availability | 8. equipment state | 9. process transfer | 10. release status

Two different limits

F Say it

Two tools have sample means of 50 and 53 nm. Both meet a mean specification of 48 to 55 nm. Matching allows at most 1 nm difference. Transfer approval is pending.

Complete the four gaps. Use each bank entry once.

06

WORD + PHRASE BANK

3 / pending / 1 / matched

Engineer: "The means differ by ¹ _____ nm."

Reviewer: "The matching limit is ² _____ nm."

Engineer: "The means are in spec, but the tools are not ³ _____."

Reviewer: "Transfer approval remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 87

1. 3 | 2. 1 | 3. matched | 4. pending

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

One completed test is not complete qualification

A Read the situation

Explain qualification status test by test, preserve configuration scope, and correct an overbroad customer claim.

Product engineer Diego and reliability engineer Mei review package Q7. Its approved fictional qualification plan requires temperature cycling, high-temperature operating life testing, a highly accelerated stress test, and final review of the evidence. Temperature cycling has passed under its protocol. Operating-life testing is running, and the accelerated stress test has not started. No final approval or accepted substitution is recorded. A customer slide nevertheless says qualification complete. The team must replace that headline with a precise status without describing open tests as failures.

PEOPLE / Diego: Product engineer | Mei: Reliability engineer

07

GIVE PROGRESS WITHOUT COLLAPSING THE PLAN

A customer may prefer a short headline, but complete must refer to the required plan and configuration. Name the finished test, the open work, and the pending decision. Do not imply that a test still running has passed, or that a test not yet started has failed.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which test has passed?

- A Temperature cycling
- B High-temperature operating life
- C The highly accelerated stress test
- D Every required test and final review

2. What is the operating-life test status?

- A Running
- B Passed and approved
- C Not required by the plan
- D Replaced by an accepted substitution

3. Why is qualification complete unsupported?

- A Every package test is optional.
- B A running test is automatically a failure.
- C The package identifier proves approval.
- D Two required tests and final review remain open.

Answers and explanations: [page 88](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

package substrate

The package structure carrying connections between die and external interfaces.

IN USE specify the package substrate

wire bonding

Forming electrical connections using fine wires.

IN USE qualify wire bonding

flip-chip attachment

Connecting a die face-down through conductive bumps or pillars.

IN USE evaluate flip-chip attachment

underfill

Material placed around interconnections to support a package assembly.

IN USE assess underfill integrity

mold compound

Encapsulating material used in many semiconductor packages.

IN USE qualify the mold compound

solder joint

A soldered electrical and mechanical connection.

IN USE inspect solder joints

BGA

Ball grid array, a package connection format using an array of solder balls.

IN USE assess BGA assembly

QFN

Quad flat no-lead, a package format with exposed connection lands.

IN USE review the QFN package

SiP

System in package, integrating multiple functions or components in one package.

IN USE evaluate a SiP design

TSV

Through-silicon via, a conductive connection passing through silicon.

IN USE characterize TSV connections

package construction

The materials and structural arrangement of a package.

IN USE identify package construction

moisture sensitivity

Susceptibility to moisture-related effects under stated handling or stress conditions.

IN USE assess moisture sensitivity

Precision check

This case uses a specific fictional plan, not a universal mandatory three-test list. One passing test supports its defined result. It does not complete the two open tests or the final review, and it does not establish every application-specific reliability claim.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

delamination

Separation between bonded material layers.

IN USE inspect for delamination

thermal cycling

Repeated temperature changes under a specified test profile.

IN USE complete thermal cycling

HTOL

High-temperature operating life, an operating stress test under defined conditions.

IN USE report HTOL status

HAST

Highly accelerated stress test, a defined accelerated humidity-related stress test.

IN USE schedule HAST

preconditioning

Defined preparation before a test to represent relevant handling or assembly stresses.

IN USE document preconditioning

qualification plan

The defined tests, evidence, scope, and approval requirements for qualification.

IN USE follow the qualification plan

qualification vehicle

The selected device or structure used to support qualification evidence.

IN USE identify the qualification vehicle

test coverage

The requirements or failure concerns addressed by the performed tests.

IN USE assess qualification test coverage

failure analysis

Investigation of failed samples and their causes.

IN USE initiate failure analysis

qualification by similarity

Use of justified related evidence under an accepted qualification approach.

IN USE assess qualification by similarity

final test

Electrical testing at the defined final product stage.

IN USE complete final test

qualification approval

Authorized acceptance of the required qualification evidence.

IN USE obtain qualification approval

Precision check

Related package evidence may sometimes support qualification through an accepted similarity rationale. No such substitution is accepted here. Similar appearance, a shared package name, or a convenient schedule cannot silently change the approved evidence requirements.

Phrases you can actually use

C Notice the language / 1 of 2

NAME THE COMPLETED RESULT

"Q7 passed temperature cycling under the stated protocol."

STATE UNSTARTED WORK

"HAST has not started."

CORRECT THE HEADLINE

"Qualification is in progress, not complete."

PRESERVE CONFIGURATION

"Which package construction does the evidence cover?"

STATE ONGOING WORK

"HTOL is still running."

KEEP THE PLAN VISIBLE

"The approved plan requires all three tests and final review."

AVOID TREATING PENDING AS FAILED

"No result is available for the unstarted test."

ASK ABOUT THE VEHICLE

"Is this the approved qualification vehicle?"

Qualified

Specify product, package, construction, scope, and approval.

Passed

Applies to a defined test result, not automatically the entire plan.

Running

A current activity status, not a completed result.

Phrases you can actually use

C Notice the language / 2 of 2

CHECK PREPARATION

"Are the required preconditioning records included?"

QUALIFY RELATED EVIDENCE

"Similarity needs an accepted technical rationale."

STATE THE APPROVAL GAP

"Final qualification approval is not recorded."

COMMIT TO THE NEXT UPDATE

"The next report will distinguish completed, running, and unstarted tests."

SEPARATE PRODUCTION TEST

"Final test is not a substitute for every reliability stress."

AVOID A SCHEDULE SHORTCUT

"A desired customer date does not remove a required test."

DESCRIBE COVERAGE

"Identify what each completed test actually addresses."

CLOSE ACCURATELY

"Keep release decisions with the authorized review process."

Not started

Does not mean failed or waived.

Similar

Needs a justified and accepted relevance argument.

Complete

Includes the required evidence review, not only laboratory scheduling.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which customer headline is supported?

- A** Q7 is fully qualified because temperature cycling passed.
- B** Q7 failed HAST because HAST has not started.
- C** Q7 is approved by similarity despite no accepted substitution.
- D** Qualification in progress: temperature cycling passed; HTOL running; HAST not started.

2. What does the passing temperature-cycle result establish?

- A** Completion of every operating-life requirement
- B** Automatic approval for all customer applications
- C** That no other failure mechanism can occur
- D** The stated test result under its protocol and scope

3. When could related evidence replace a required test in this case?

- A** Whenever a salesperson prefers an earlier date
- B** Whenever two packages look similar in a photograph
- C** Automatically after any one test passes
- D** Only through the applicable accepted change or similarity process, which has not occurred

4. What should final review examine?

- A** Required evidence, scope, configuration, open items, and approval conditions
- B** Only whether the slide uses the word complete
- C** Only the date the first test started
- D** A presumed pass for all tests without final results

Answers and explanations: [page 88](#).

One completed test is not complete qualification

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

thermal cycling / preconditioning / qualification plan / Test coverage / qualification approval / HAST / HTOL / Qualification by similarity / qualification vehicle / package construction

07

- 01 **Diego:** The customer slide says Q7 qualification is complete because temperature cycling passed. We still have two other tests in the plan, so I need a more accurate headline.
- 02 **Mei:** Start from the ¹ _____. Our approved plan requires all three tests and final evidence review. We have completed one part, not the entire qualification decision.
- 03 **Diego:** The completed test followed its protocol and passed. I want to preserve that positive result rather than make the update sound as though nothing has been achieved.
- 04 **Mei:** Report the ² _____ result clearly. Its value is not diminished by stating the other work accurately, and it should not be expanded beyond its protocol and scope.
- 05 **Diego:** The operating-life test is still running. The lab has not supplied its final result, so the customer should not read the current status as a pass.
- 06 **Mei:** Label ³ _____ as running. If a progress checkpoint is available, identify it as interim information without treating it as the completed test outcome.
- 07 **Diego:** The accelerated humidity-related stress test has not started. That is a schedule and evidence gap, but it is not a failed result.
- 08 **Mei:** Exactly. ⁴ _____ remains unstated in this plan. Use that status rather than either a reassuring green result or an unsupported failure label.
- 09 **Diego:** The customer also needs to know which package construction the results cover. A broad Q7 name could hide differences in materials or assembly details.
- 10 **Mei:** Keep the ⁵ _____ linked to the evidence. Qualification scope should not silently expand to a different variant simply because the marketing name is similar.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 64.

- 11 **Diego:** We should confirm the tested device or structure as well. Otherwise reviewers may not know how the selected samples relate to the configuration being discussed.
- 12 **Mei:** Identify the ⁶ _____ and its approved rationale. Traceability makes it possible to review applicability rather than assume that any available sample represents every product.
- 13 **Diego:** The assembly preparation records also belong in the package. I do not want the final review to discover that important preparation details were omitted from the summary.
- 14 **Mei:** Include the required ⁷ _____ records where the plan calls for them. A passing result needs the relevant preparation and test context to be interpreted correctly.
- 15 **Diego:** Sales asked whether an older package result could stand in for the tests still open. No substitution has been reviewed or accepted for this plan.
- 16 **Mei:** ⁸ _____ needs an accepted basis under the applicable process. A related package or a desired delivery date does not automatically change the approved requirements.
- 17 **Diego:** We also have final electrical test records, but those should not be presented as though they address every stress-related concern in the qualification plan.
- 18 **Mei:** Correct. ⁹ _____ differs between production checks and reliability assessments. Each result needs to be connected to what it actually evaluates, not treated as interchangeable evidence.
- 19 **Diego:** I will change the headline to qualification in progress, list the three statuses, and keep the configuration and outstanding review visible. We will not announce release from this slide.
- 20 **Mei:** That preserves ¹⁰ _____ as a separate decision. The next update can show genuine progress while keeping the unresolved tests and final review clear.

07

ANSWERS / EXPLANATIONS: PAGE 89

1. qualification plan | 2. thermal cycling | 3. HTOL | 4. HAST | 5. package construction | 6. qualification vehicle | 7. preconditioning | 8. Qualification by similarity | 9. Test coverage | 10. qualification approval

Pending is neither passed nor failed

F Say it

A fictional package plan requires tests A, B, and C plus final review. A passed, B is running, and C has not started. No substitution or final approval is recorded.

Complete the four gaps. Use each bank entry once.

07

WORD + PHRASE BANK

started / pending / running / passed

Engineer: "Test A has ¹ _____."

Reviewer: "Test B is ² _____."

Engineer: "Test C has not ³ _____."

Reviewer: "Qualification approval remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 89

1. passed | 2. running | 3. started | 4. pending

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A possible slot is not tape-out readiness

A Read the situation

Negotiate a requested date by separating design sign-off, data delivery, foundry acceptance, and capacity confirmation.

Customer design lead Sofia asks foundry liaison Ken to move a planned October 15 tape-out to October 8. Four design-rule-check findings and one layout-versus-schematic mismatch remain unresolved in the agreed fictional PDK revision P3. The foundry has mentioned a possible earlier slot but has not reserved or accepted it. No complete sign-off package is available, and no waiver is approved. Ken must communicate the blockers and conditional next steps without promising a wafer start, finished samples, or shipment.

PEOPLE / Sofia: Customer design lead | Ken: Foundry liaison

08

SEPARATE THE MILESTONES UNDER DATE PRESSURE

Tape-out, foundry data acceptance, wafer start, and finished samples are related but distinct milestones. State which date is requested and which prerequisites remain open. A possible capacity opportunity should trigger a feasibility discussion, not erase design checks or become a shipment promise.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which earlier date is requested?

A October 15 completed wafer sort

B October 8 tape-out

C October 8 finished samples

D October 15 shipment

2. What remains unresolved in the design checks?

A An approved waiver for all findings

B Four DRC findings and one LVS mismatch

C A completed sign-off package with no open items

D Only a customer shipping address

3. What is the capacity status?

A A confirmed wafer start

B A guaranteed sample delivery

C An approved bypass of design sign-off

D A possible earlier slot, not reserved or accepted

Answers and explanations: page 90.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

foundry

A manufacturer producing semiconductor devices for other companies.

IN USE coordinate with the foundry

fabless company

A chip-design company that contracts out fabrication.

IN USE support a fabless customer

IDM

Integrated device manufacturer, combining chip design and manufacturing.

IN USE work with an IDM

PDK

Process design kit: process-specific models, rules, and design resources.

IN USE confirm the PDK revision

design rule

A process-specific constraint governing layout features or relationships.

IN USE check design rules

DRC

Design rule checking, testing layout against applicable rule definitions.

IN USE resolve DRC findings

LVS

Layout versus schematic checking, comparing extracted layout connectivity with the schematic.

IN USE close an LVS mismatch

rule deck

The implemented rules used by a physical-verification tool.

IN USE identify the rule deck

sign-off

The required formal completion and approval of defined checks.

IN USE complete design sign-off

tape-out

Release of the approved design dataset for the intended fabrication handoff.

IN USE plan the tape-out

GDSII

A widely used layout-data format for integrated circuit design.

IN USE deliver a GDSII dataset

OASIS

A compact integrated-circuit layout interchange format.

IN USE verify the OASIS file

Precision check

October 8 is a requested tape-out date, not an accepted slot or a finished-sample date. The open DRC and LVS items remain blockers to the stated sign-off package. The case does not provide an approved waiver or permission to bypass checks.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

mask data preparation

Processing layout data for mask manufacture.

IN USE schedule mask data preparation

mask set

The collection of masks needed for a defined fabrication flow.

IN USE release the mask set

engineering change order

A controlled record authorizing a defined engineering change, abbreviated ECO.

IN USE approve an engineering change order

waiver

An authorized exception to a specific requirement under an applicable process.

IN USE request a defined waiver

data acceptance

Confirmation that a delivered dataset meets the receiving process's requirements.

IN USE obtain foundry data acceptance

capacity reservation

A confirmed allocation of manufacturing capacity.

IN USE confirm capacity reservation

wafer start

Entry of a wafer lot into the defined fabrication flow.

IN USE confirm the wafer start

cycle time

Elapsed time through a defined manufacturing flow or stage.

IN USE state fabrication cycle time

engineering lot

A lot used for development or engineering evaluation.

IN USE plan an engineering lot

risk production

Production begun before all defined qualification or maturity conditions are complete.

IN USE assess risk-production approval

respin

A revised design iteration requiring another fabrication cycle.

IN USE evaluate the need for a respin

delivery commitment

An agreed promise about a specified deliverable and date.

IN USE qualify the delivery commitment

Precision check

PDK and rule-deck versions affect what physical verification means. Keep results connected to the agreed revision. A clean check run also has a defined scope; it does not guarantee first-silicon performance, complete qualification, or a confirmed manufacturing schedule.

Phrases you can actually use

C Notice the language / 1 of 2

NAME THE REQUESTED MILESTONE

"You are requesting October 8 tape-out."

STATE THE OPEN CHECKS

"Four DRC findings and one LVS mismatch remain open."

AVOID SILENT VERSION CHANGES

"Keep the rule deck and design dataset identified."

ASK ABOUT ACCEPTANCE

"What is required for the foundry to accept the dataset?"

SEPARATE LATER DATES

"That is not a wafer-start or sample-delivery commitment."

CONFIRM THE BASIS

"The checks use the agreed PDK revision P3."

QUALIFY THE SLOT

"The foundry mentioned a possibility, not a reservation."

KEEP SIGN-OFF VISIBLE

"The complete sign-off package is not yet available."

08

Slot

Clarify whether possible, held, reserved, or confirmed.

Ready

Identify which handoff and its required checks.

Tape-out

Not interchangeable with wafer start or finished-device delivery.

Phrases you can actually use

C Notice the language / 2 of 2

BOUND THE EXCEPTION

"No waiver is approved for these findings."

REQUEST A FEASIBLE SEQUENCE

"Align check closure, data review, and capacity confirmation."

CLARIFY OWNERSHIP

"Who owns each open check and the next status update?"

STATE THE NEXT UPDATE

"We will report check status and slot confirmation separately."

PRESERVE THE DECISION ROUTE

"Any exception must follow the applicable authorization process."

AVOID A DOWNSTREAM PROMISE

"We cannot infer finished samples from a tape-out target."

RECORD THE DEPENDENCY

"The earlier request depends on both technical readiness and capacity."

CLOSE CONDITIONALLY

"October 8 remains a request pending the required evidence and acceptance."

08

Clean

State the check scope, version, and remaining exceptions.

Waived

Requires an actual authorized exception, not a schedule preference.

Committed

Name the deliverable, date, dependencies, and accepting party.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which reply to the earlier-date request is supported?

- A** October 8 remains conditional on check closure, the required handoff, and capacity confirmation.
- B** October 8 guarantees finished samples because a slot was mentioned.
- C** The unresolved findings disappear if the customer accepts schedule risk.
- D** The foundry has already reserved capacity by discussing it.

08

2. What does an LVS mismatch concern?

- A** A difference between extracted layout connectivity and the schematic
- B** Only an available tool's uptime percentage
- C** The package's shipping label
- D** The number of wafers already shipped

3. Why retain the PDK and rule-deck versions with results?

- A** A newer version is automatically approved for this project.
- B** The check result depends on the process-specific definitions actually used.
- C** A filename guarantees every future revision is identical.
- D** Version records eliminate every physical verification requirement.

4. Which statement about tape-out is accurate?

- A** It always occurs on the same day as customer shipment.
- B** It guarantees that no respin can be needed.
- C** It is a design-data handoff milestone, not proof of wafer start or sample availability.
- D** It means every package has completed reliability qualification.

Answers and explanations: page 90.

A possible slot is not tape-out readiness

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

delivery commitment / tape-out / sign-off / DRC / LVS / capacity reservation / waiver / PDK / Data acceptance / wafer start

08

- 01 **Sofia:** Can we move tape-out from October fifteenth to October eighth? The foundry mentioned an earlier slot, and the customer is asking whether that means samples can arrive sooner.
- 02 **Ken:** Separate the ¹ _____ request from the later milestones. An earlier data handoff may matter to the plan, but it does not itself establish a wafer start or sample-delivery date.
- 03 **Sofia:** We still have four design-rule findings and one layout-versus-schematic mismatch. I want to know whether those can remain open while we secure the opportunity.
- 04 **Ken:** The ² _____ package is not complete. No waiver is approved, so we cannot describe the design as ready by treating the open checks as an informal schedule exception.
- 05 **Sofia:** The verification team is working against P3, the agreed kit revision. We should preserve that basis when reporting progress rather than mix results from different revisions.
- 06 **Ken:** Keep the ³ _____ revision with the dataset and check results. The design rules and supported flow are process-specific, so version identity is part of the evidence.
- 07 **Sofia:** The four rule findings need owners and disposition. A count alone does not show whether they are being resolved or merely reclassified.
- 08 **Ken:** Report each ⁴ _____ finding with its status under the agreed process. Do not call the run clean while unresolved findings or required decisions remain outside the summary.
- 09 **Sofia:** The connectivity mismatch is a separate issue. I do not want the customer to hear that all five items are the same kind of geometric spacing check.
- 10 **Ken:** Correct. ⁵ _____ addresses layout versus schematic consistency. Keep that distinction clear when assigning ownership and explaining why the complete sign-off package is not yet available.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 73.

- 11 **Sofia:** On capacity, the foundry said a slot might become available. We have no written confirmation that it is allocated to this design.
- 12 **Ken:** Then there is no confirmed ⁶ _____. We can discuss feasibility, but a mentioned opportunity must not be reported as an accepted manufacturing commitment.
- 13 **Sofia:** Even after we complete the design package, the receiving team will need to check the delivered data against its requirements. Sending files is not necessarily the same as acceptance.
- 14 **Ken:** Exactly. ⁷ _____ is a separate handoff status. Record what the foundry needs and who confirms it, rather than assume a sent dataset has cleared the receiving review.
- 15 **Sofia:** Could a formal exception be considered if one finding remains? I want to ask through the correct route, not imply that my request already authorizes it.
- 16 **Ken:** A ⁸ _____ would need the applicable technical assessment and authorization. We cannot assume it is available or accepted simply because the earlier date is commercially attractive.
- 17 **Sofia:** I will avoid calling October eighth a fabrication start. The manufacturing schedule also depends on accepted data and actual capacity arrangements beyond our requested handoff.
- 18 **Ken:** Correct. A ⁹ _____ is a different milestone. Its timing and the subsequent flow must be confirmed on their own basis before anyone promises finished samples.
- 19 **Sofia:** The next update will separate check closure from slot confirmation. It will identify owners and show what remains conditional rather than announce the earlier date as secured.
- 20 **Ken:** That protects the ¹⁰ _____. We can pursue the opportunity actively while making clear which evidence, acceptance, and manufacturing decisions still stand between a request and a confirmed outcome.

08

ANSWERS / EXPLANATIONS: PAGE 91

1. tape-out | 2. sign-off | 3. PDK | 4. DRC | 5. LVS | 6. capacity reservation | 7. Data acceptance | 8. waiver | 9. wafer start | 10. delivery commitment

A requested handoff is not a shipment

F Say it

A team requests tape-out on November 3. Two DRC items remain open. The foundry has mentioned a possible slot but has not reserved it. No waiver or sample-delivery date is approved.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

approved / tape-out / reserved / open

08

Designer: "November 3 is the requested ¹ _____ date."

Liaison: "The two DRC items remain ² _____."

Designer: "The possible slot is not ³ _____."

Liaison: "No sample-delivery date is ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 91

1. tape-out | 2. open | 3. reserved | 4. approved

Rehearse with a fixed script

1. Check your answers, then read turns 1-10 with a partner using the corrected words.
2. Switch roles for turns 11-20. Stress words that distinguish facts from conditions or estimates.
3. Complete and check the four-line exchange. Read it twice, switching roles and keeping the printed facts.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

01 Briefing and language checks

Wafer Fabrication Flow and Process Integration

A Briefing checks / page 4

1. Where is L24 now?

D. In wafer fabrication at an M2-related operation

The briefing places L24 in wafer fabrication, not the packaging stage Eric inferred.

2. What does BEOL mean in this case?

A. Back end of line, forming on-wafer interconnects

The integration engineer uses BEOL for on-wafer interconnect fabrication.

3. What prevents the next operation?

C. A pending inspection review and lot hold

The lot remains on hold until the stated inspection review and authorized disposition.

D Language checks / page 9

1. Which customer update is accurate?

A. L24 remains in wafer fabrication at M2, on hold for inspection review; packaging has not started.

The update preserves the actual stage and hold without inventing later progress.

2. Why expand BEOL for the coordinator?

D. Different teams can use back-end for different parts of the manufacturing chain.

Expanding the term prevents a cross-team ambiguity that already caused an inaccurate update.

3. What is the appropriate next-step wording?

B. The lot awaits inspection review and authorized disposition under its route.

The stated hold and route control what may happen next, not a generic assumption about progress.

4. Which distinction is correct?

A. Wafer sort tests die on the wafer; package assembly integrates die into packages.

The terms describe different activities and cannot be used as interchangeable progress labels.

01 Conversation answers

Wafer Fabrication Flow and Process Integration

E Dialogue gaps

Pages 10-11

1. BEOL. Turn 2. BEOL refers to the on-wafer interconnect stage in this case, not to completed package assembly.

2. wafer fabrication. Turn 4. Wafer fabrication locates the current work on the wafer before the package assembly stage described in the mistaken update.

3. metal layer. Turn 6. The metal layer identifies the specific interconnect level involved in the current operation.

4. FEOL. Turn 8. FEOL principally forms active devices, distinct from the later on-wafer interconnect work now involving L24.

5. lot hold. Turn 10. A lot hold restricts further movement until the required review and authorized disposition occur.

6. disposition. Turn 12. Disposition is the authorized decision about how the lot may proceed after review, not an assumption based on its position in the flow.

7. traveler. Turn 14. The traveler connects the material with its operation history and required next steps, supporting traceable status communication.

8. Wafer sort. Turn 16. Wafer sort is electrical testing of die on the wafer and is distinct from package assembly or shipment.

9. package assembly. Turn 18. Package assembly is the later integration of die into packages, which has not started for L24.

10. process flow. Turn 20. The process flow connects the current stage and authorized next steps without collapsing fabrication, testing, and assembly into one label.

F Transfer conversation

Page 12

1. fabrication

Coordinator: "The lot is still in wafer fabrication."

Forming on-wafer interconnects belongs to wafer fabrication in this case.

2. hold

Engineer: "Its current restriction is a lot hold."

The briefing explicitly states that the lot is on hold for review.

3. started

Coordinator: "Package assembly has not started."

The supplied facts say package assembly has not yet begun.

4. unconfirmed

Engineer: "The shipping date remains unconfirmed."

No confirmed shipping date is supplied for the remaining process flow.

02 Briefing and language checks

Lithography, Reticles, and Critical Dimensions

A Briefing checks / page 13

1. Which layer and stage do the records identify?

D. M2 after etch

The recovered records specifically locate the two means on M2 after etch.

2. What is the reported numerical difference?

C. An increase of 3 nm

Subtracting the first mean, 86 nm, from the second, 89 nm, gives a 3 nm increase.

3. Why is a direct process attribution premature?

C. The measurement recipes and sampling sites differ.

Changed measurement and sampling conditions prevent the difference from isolating a process cause.

D Language checks / page 18

1. Which report sentence is justified?

C. The means differ by 3 nm, but recipe and site changes limit comparability.

The sentence reports the arithmetic while preserving the unresolved measurement basis.

2. Which comparison best helps investigate the apparent change?

C. Comparable measurements on identified sites using an agreed recipe basis

Comparable measurement conditions help distinguish process behavior from measurement and sampling differences.

3. Why is post-etch important?

B. The result follows pattern transfer and does not isolate lithography alone.

The measurement stage affects interpretation because more than the exposure step contributes to the resulting feature.

4. Which distinction is accurate?

A. CD concerns feature size; overlay concerns layer alignment.

Feature size and alignment describe different pattern properties and require distinct measurements.

02 Conversation answers

Lithography, Reticles, and Critical Dimensions

E Dialogue gaps

Pages 19-20

- 1. post-etch.** Turn 2. Post-etch locates the measurement after pattern transfer, which matters when interpreting possible causes.
- 2. measurement recipe.** Turn 4. The measurement recipe defines settings and analysis that can affect the reported CD and differs between these datasets.
- 3. sampling site.** Turn 6. A sampling site is a selected measurement location; different site selections can affect the reported average.
- 4. critical dimension.** Turn 8. Critical dimension is the relevant feature size, but its reported change does not alone establish the physical cause.
- 5. Etch bias.** Turn 10. Etch bias concerns the dimensional change associated with pattern transfer and is one reason post-etch CD does not isolate exposure.
- 6. Overlay.** Turn 12. Overlay describes alignment between layers, whereas CD describes a feature dimension.
- 7. remeasurement.** Turn 14. Remeasurement on an agreed basis can help determine whether the apparent difference persists under comparable conditions.
- 8. measurement bias.** Turn 16. Measurement bias is a systematic difference in measurement relative to a reference and is distinct from an assumed process shift.
- 9. reticle.** Turn 18. The reticle carries the lithographic pattern; identifying its revision provides context without proving it caused the observation.
- 10. pattern fidelity.** Turn 20. Pattern fidelity concerns how well the produced pattern matches its intent; assessing it requires more than an unexplained mean difference.

F Transfer conversation

Page 21

1. 4

Engineer: "The apparent increase is 4 nm."

Subtracting 120 from 124 gives a four-nanometer numerical increase.

2. etch

Reviewer: "The measurements were taken after etch."

The briefing explicitly identifies both values as post-etch measurements.

3. different

Engineer: "The measurement recipes were different."

The two datasets do not share the same stated measurement recipe.

4. unconfirmed

Reviewer: "The cause remains unconfirmed."

Changed measurement and sampling conditions prevent a confirmed causal interpretation here.

03 Briefing and language checks

Deposition, Etch, CMP, and Process Windows

03

A Briefing checks / page 22

1. What improved in the supplied trial?

B. The stated nonuniformity metric fell from 5% to 3% at the nominal setting.

The only reported improvement concerns the unchanged metric at one tested condition.

2. What is the absolute change in that percentage metric?

A. A decrease of 2 percentage points

Five percent minus three percent equals two percentage points on the stated metric.

3. Which conclusion is premature?

C. The complete process window is validated.

The broader range and downstream behavior were not assessed, so full-window validation is unsupported.

D Language checks / page 27

1. Which summary preserves the evidence boundary?

B. The nominal-setting metric improved; the range and downstream effects remain untested.

The statement separates the observed response from the untested range and integration behavior.

2. What is the relative reduction from 5% to 3%?

C. 40%

The decrease of two divided by the original five equals forty percent.

3. Why review downstream etch and CMP?

B. A film change may alter later process behavior.

Connected process steps can create trade-offs not captured by one deposition response.

4. Which status should remain separate from trial success?

B. Recipe release

Recipe release is an authorized decision that has not occurred in this case.

03 Conversation answers

Deposition, Etch, CMP, and Process Windows

03

E Dialogue gaps

Pages 28-29

1. process window. Turn 2. A process window is a range satisfying defined requirements, not a single successful nominal setting.

2. thickness nonuniformity. Turn 4. Thickness nonuniformity is the specific response that improved; it does not stand for all film or integration properties.

3. nominal setting. Turn 6. The nominal setting is the only process condition tested, so the measured improvement must remain limited to it.

4. corner condition. Turn 8. A corner condition is a defined limiting setting or combination that requires its own evidence in the evaluation plan.

5. integration trade-off. Turn 10. An integration trade-off concerns benefits and consequences across related steps that a single improved deposition metric cannot settle.

6. etch selectivity. Turn 12. Etch selectivity compares material removal rates and may be affected by a film change; no result is supplied here.

7. CMP. Turn 14. CMP is chemical mechanical planarization, a downstream process whose compatibility with the changed film remains unassessed.

8. process interaction. Turn 16. A process interaction is a relationship between factors or steps that can change the overall result and needs explicit assessment.

9. recipe release. Turn 18. Recipe release is the controlled authorization for defined use, separate from a promising experimental result.

10. deposition. Turn 20. Deposition is the film-forming step actually assessed; the statement should not imply the entire integrated flow was validated.

F Transfer conversation

Page 30

1. 2

Engineer: "The decrease is 2 percentage points."

Eight percent minus six percent is a two-percentage-point decrease.

2. 25

Reviewer: "The relative reduction is 25 percent."

Two divided by the original eight equals a twenty-five percent reduction.

3. nominal

Engineer: "Only the nominal setting was tested."

The briefing limits the available result to the nominal setting.

4. release

Reviewer: "The recipe still needs release."

The recipe has not received release despite the favorable trial metric.

04 Briefing and language checks

Metrology, SPC, and Yield Learning

04

A Briefing checks / page 31

1. What is the pilot yield?

B. 92% from 92 of 100 tested die

Ninety-two passing die divided by one hundred tested gives a 92% observed yield.

2. What is the absolute difference between the observed yields?

C. 2 percentage points

The observed yields are 92% and 90%, a difference of two percentage points.

3. Which factor limits the causal comparison?

C. The product mixes differ.

Different product mixes can affect yield, so the process change is not isolated by this comparison.

D Language checks / page 36

1. Which announcement is supported?

D. The pilot observed 92% versus 90% yield, with different product mixes and no causal conclusion.

The counts support descriptive percentages, but the comparison does not isolate cause or establish stability.

2. What is the approximate relative increase from 90% to 92%?

B. 2.22%

The two-point increase divided by the 90% starting value is approximately 2.22%.

3. Which next analysis addresses the product-mix problem?

B. Separate comparable product groups and review consistent test bases.

Stratification helps compare relevant groups rather than allowing product composition to drive the headline.

4. Which statement distinguishes control and specification limits?

A. Control limits monitor behavior; specification limits define acceptance boundaries.

Their purposes differ, and investigation and disposition require the applicable evidence and procedures.

04 Conversation answers

Metrology, SPC, and Yield Learning

E Dialogue gaps

Pages 37-38

- 1. pass count.** Turn 2. The pass count identifies the actual successes behind the percentage, preserving the scale of the evidence.
- 2. product mix.** Turn 4. Product mix differs between the lots and may affect observed yield independently of the process change.
- 3. percentage point.** Turn 6. Percentage point measures the absolute difference between the two yield percentages, here 92 minus 90.
- 4. causal attribution.** Turn 8. Causal attribution assigns the observed yield difference to the process change, which this mixed comparison does not isolate.
- 5. stratification.** Turn 10. Stratification separates the data into comparable product groups to examine the mix effect more directly.
- 6. process stability.** Turn 12. Process stability concerns statistical behavior over time under a defined monitoring method, not two favorable aggregate yields.
- 7. specification limit.** Turn 14. A specification limit defines acceptance requirements and is distinct from a statistically based process-monitoring boundary.
- 8. control signal.** Turn 16. A control signal triggers investigation under the monitoring method; it does not provide its own causal diagnosis.
- 9. measurement system.** Turn 18. The measurement system includes methods and conditions that affect whether the collected results can be compared consistently.
- 10. yield learning.** Turn 20. Yield learning uses the observations to investigate losses and drivers without claiming an effect before the evidence supports it.

F Transfer conversation

Page 39

1. 50

Engineer: "The trial denominator is 50."

Fifty die were tested in the trial, which supplies its denominator.

2. 500

Reviewer: "The reference denominator is 500."

Five hundred die were tested in the reference dataset.

3. 6

Engineer: "The absolute yield difference is 6 percentage points."

Ninety-six percent minus ninety percent equals six percentage points.

4. unproven

Reviewer: "A causal process improvement remains unproven."

Different product mixes and incomplete analysis prevent a proven causal conclusion.

05 Briefing and language checks

Defect Density and Cleanroom Contamination

A Briefing checks / page 40

1. What does the inspection record contain?

A. 24 particle-like detections over 200 square centimeters

The record describes surface inspection detections and inspected area, not confirmed failure or added-defect counts.

2. What does the map show?

D. An edge cluster

The supplied observation is a spatial edge cluster, without an established source.

3. Why cannot the 24 detections be called 24 added defects?

C. No comparable pre-process scan exists, and review is incomplete.

Added defects require a valid before-and-after basis, and the detections are not all confirmed defects.

D Language checks / page 45

1. What is the observed detection density?

B. 0.12 detections per square centimeter

Twenty-four detections divided by two hundred square centimeters equals 0.12 on a surface-area basis.

2. Which conclusion is unsupported?

D. C2 introduced all 24 detections.

Timing after C2 does not establish introduction by C2 without the needed comparison and investigation.

3. Which next record is useful for the investigation?

C. Traceable wafer, inspection-setting, timing, and exposure information

Relevant process and inspection context helps assess the spatial pattern without substituting blame for evidence.

4. Which distinction is correct?

A. Surface detection density and airborne particle counts use different measurement bases.

Surface area and air volume describe different measurements and cannot be silently substituted.

05 Conversation answers

Defect Density and Cleanroom Contamination

E Dialogue gaps

Pages 46-47

1. inspected area. Turn 2. The inspected area provides the denominator for the surface detection density rather than leaving the count without scale.

2. Defect review. Turn 4. Defect review determines how flagged features should be classified before they are treated as confirmed relevant defects.

3. edge cluster. Turn 6. An edge cluster describes the spatial concentration actually observed without claiming its origin has been established.

4. added defect. Turn 8. An added defect requires evidence that it appeared between comparable inspection points, which is missing here.

5. pre-process scan. Turn 10. A pre-process scan provides the earlier observation needed for a valid before-and-after comparison when methods are comparable.

6. airborne particle count. Turn 12. An airborne particle count concerns a defined air sample or volume, unlike the surface-area-based detections here.

7. inspection sensitivity. Turn 14. Inspection sensitivity affects which features are detected and must be considered when comparing inspection results.

8. containment scope. Turn 16. Containment scope identifies which material or operations are restricted while the investigation remains open.

9. exposure history. Turn 18. Exposure history connects the wafers to relevant handling and environments that may help evaluate possible contamination origins.

10. contamination source. Turn 20. The contamination source is the origin to be established by investigation, not assumed from the latest process step or vague label.

F Transfer conversation

Page 48

1. 0.12

Engineer: "The observed density is 0.12 detections per square centimeter."

Twelve flagged features divided by one hundred square centimeters equals 0.12.

2. pending

Reviewer: "Feature classification is still pending."

The briefing states that classification has not yet been completed.

3. pre-process

Engineer: "There is no comparable pre-process scan."

The missing earlier scan prevents a supported before-and-after attribution.

4. unconfirmed

Reviewer: "The contamination source is unconfirmed."

The supplied observations do not establish the origin of the flagged features.

05

06 Briefing and language checks

Equipment Uptime, Recipes, and Tool Matching

A Briefing checks / page 49

1. How far apart are the sample means?

D. 4 nm

The difference between 102 nm and 98 nm is four nanometers.

2. Do the means meet the stated matching criterion?

C. No; 4 nm exceeds the allowed 2 nm difference.

The matching criterion applies to the difference between means, not merely each mean's specification status.

3. What does 90% describe here?

A. Availability for 900 of the defined 1,000 minutes

The supplied time ratio concerns equipment availability and establishes no product-yield or approval claim.

D Language checks / page 54

1. Which status report is accurate?

C. Both means meet the mean specification; their 4 nm difference fails the 2 nm matching limit.

The specification and matching tests use different criteria, and individual results are not supplied.

2. What does using the same recipe establish?

B. The identified recipe input is shared, not that output performance is interchangeable.

Recipe identity alone does not demonstrate equivalent physical conditions or process outcomes.

3. Which calculation matches the availability basis?

C. 900 divided by 1,000 equals 90%.

Availability uses the supplied available time divided by the defined total window.

4. What remains necessary before calling B an approved interchangeable backup?

B. The required matching or qualification evidence and transfer approval

The case lacks matching acceptance and transfer approval, neither of which is replaced by uptime.

06 Conversation answers

Equipment Uptime, Recipes, and Tool Matching

E Dialogue gaps

Pages 55-56

1. Tool matching. Turn 2. Tool matching requires measured performance against a defined comparison criterion, not simply a shared recipe or uptime figure.

2. matching criterion. Turn 4. The matching criterion allows no more than two nanometers between means; the observed difference is four.

3. actual value. Turn 6. Actual value concerns the measured outcome; passing the mean specification does not remove the separate difference between tools.

4. recipe revision. Turn 8. The recipe revision identifies the shared controlled input, but cannot stand in for unsupplied individual wafer evidence.

5. chamber condition. Turn 10. Chamber condition is a possible contributor to tool behavior that must be assessed rather than assumed from the recipe.

6. calibration status. Turn 12. Calibration status concerns the documented measurement basis and is relevant to interpreting the observed tool difference.

7. availability. Turn 14. Availability is the available-time fraction on the stated basis, not a measure of matching, yield, or approval.

8. equipment state. Turn 16. Equipment state records the tool's operational condition; an available state does not establish qualified process performance.

9. process transfer. Turn 18. Process transfer moves a defined process to another tool and requires its own relevant assessment and approval.

10. release status. Turn 20. Release status records authorization for the intended use, which remains absent despite the tool's availability and passing means.

F Transfer conversation

Page 57

1. 3

Engineer: "The means differ by 3 nm."

Fifty-three minus fifty gives a three-nanometer difference between means.

2. 1

Reviewer: "The matching limit is 1 nm."

The supplied matching criterion allows no more than one nanometer difference.

3. matched

Engineer: "The means are in spec, but the tools are not matched."

The three-nanometer difference exceeds the separate one-nanometer matching limit.

4. pending

Reviewer: "Transfer approval remains pending."

The briefing states that the transfer decision has not been approved.

07 Briefing and language checks

Packaging, Test, and Reliability Qualification

A Briefing checks / page 58

1. Which test has passed?

A. Temperature cycling

The briefing identifies temperature cycling as the only completed passing test.

2. What is the operating-life test status?

A. Running

The supplied plan still includes the test, and its stated status is running.

3. Why is qualification complete unsupported?

D. Two required tests and final review remain open.

The defined plan requires evidence and final review that have not all been completed.

D Language checks / page 63

1. Which customer headline is supported?

D. Qualification in progress: temperature cycling passed; HTOL running; HAST not started.

The headline accurately separates each test's status without inventing a result or approval.

2. What does the passing temperature-cycle result establish?

D. The stated test result under its protocol and scope

A test result supports the claim within its defined conditions rather than universal qualification.

3. When could related evidence replace a required test in this case?

D. Only through the applicable accepted change or similarity process, which has not occurred

The plan cannot be silently changed; an applicable authorized route and accepted rationale would be needed.

4. What should final review examine?

A. Required evidence, scope, configuration, open items, and approval conditions

Qualification approval depends on the required evidence and scope, not a favorable summary label.

07 Conversation answers

Packaging, Test, and Reliability Qualification

E Dialogue gaps

Pages 64-65

1. qualification plan. Turn 2. The qualification plan defines all required tests and final review, so one passing test cannot complete it.

2. thermal cycling. Turn 4. Thermal cycling is the completed temperature-change stress test; its passing result remains valid within its defined protocol.

3. HTOL. Turn 6. HTOL is high-temperature operating life testing, which is currently running rather than completed in this case.

4. HAST. Turn 8. HAST is the highly accelerated stress test that has not started; absence of a result is not a pass or failure.

5. package construction. Turn 10. Package construction identifies the materials and structure covered by the evidence, preventing unsupported extension to other variants.

6. qualification vehicle. Turn 12. The qualification vehicle is the selected device or structure whose results support the defined qualification scope.

7. preconditioning. Turn 14. Preconditioning records show the defined preparation used before testing and form part of interpreting the evidence.

8. Qualification by similarity. Turn 16. Qualification by similarity requires an accepted technical rationale; no such substitution is established in this case.

9. Test coverage. Turn 18. Test coverage identifies which requirements or concerns the available tests address; final electrical testing does not replace every reliability stress.

10. qualification approval. Turn 20. Qualification approval is the authorized acceptance of the required evidence, which remains pending while tests and review are incomplete.

F Transfer conversation

Page 66

1. passed

Engineer: "Test A has passed."

The supplied plan status identifies A as the completed passing test.

2. running

Reviewer: "Test B is running."

B is in progress and does not yet have a final result.

3. started

Engineer: "Test C has not started."

The briefing states that C has not yet begun.

4. pending

Reviewer: "Qualification approval remains pending."

Required tests and final review are incomplete, and no approval is recorded.

08 Briefing and language checks

Foundry, Tape-Out, PDK, and Capacity Communication

A Briefing checks / page 67

1. Which earlier date is requested?

B. October 8 tape-out

Sofia asks to move tape-out from October 15 to October 8, not to deliver finished devices then.

2. What remains unresolved in the design checks?

B. Four DRC findings and one LVS mismatch

The briefing explicitly lists four design-rule findings and one layout-versus-schematic mismatch.

3. What is the capacity status?

D. A possible earlier slot, not reserved or accepted

A mentioned opportunity is not a recorded capacity commitment or approval.

D Language checks / page 72

1. Which reply to the earlier-date request is supported?

A. October 8 remains conditional on check closure, the required handoff, and capacity confirmation.

The reply preserves the separate technical and capacity prerequisites without inventing acceptance.

2. What does an LVS mismatch concern?

A. A difference between extracted layout connectivity and the schematic

LVS checks layout against schematic information, so its mismatch is a design-verification issue.

3. Why retain the PDK and rule-deck versions with results?

B. The check result depends on the process-specific definitions actually used.

Traceable versions show which rules and models underlie the reported verification status.

4. Which statement about tape-out is accurate?

C. It is a design-data handoff milestone, not proof of wafer start or sample availability.

Fabrication, test, qualification, and delivery remain distinct activities after the design-data handoff.

08 Conversation answers

Foundry, Tape-Out, PDK, and Capacity Communication

E Dialogue gaps

Pages 73-74

1. tape-out. Turn 2. Tape-out is the design-data release milestone being requested, not a confirmed wafer start or finished-sample date.

2. sign-off. Turn 4. Sign-off is the required completion and approval of the defined checks; the complete package is not yet available here.

3. PDK. Turn 6. The PDK supplies process-specific design resources; its agreed revision must remain connected to the verification results.

4. DRC. Turn 8. DRC checks the layout against applicable design rules; its open findings need documented resolution rather than a changed headline.

5. LVS. Turn 10. LVS compares extracted layout connectivity with the schematic and therefore addresses a different verification question from geometric rule checks.

6. capacity reservation. Turn 12. Capacity reservation is an actual allocation; a possible earlier slot has not been reserved or accepted in this case.

7. Data acceptance. Turn 14. Data acceptance confirms the receiving process's requirements have been met; transmitting files alone does not establish it.

8. waiver. Turn 16. A waiver is an authorized exception to a specific requirement; none is approved for the open findings here.

9. wafer start. Turn 18. A wafer start is the lot's entry into fabrication and is distinct from the earlier design-data handoff milestone.

10. delivery commitment. Turn 20. A delivery commitment must identify an accepted deliverable and date; neither a requested tape-out nor a possible slot guarantees finished samples.

F Transfer conversation

Page 75

1. tape-out

Designer: "November 3 is the requested tape-out date."

The briefing identifies November 3 as the requested design-data handoff.

2. open

Liaison: "The two DRC items remain open."

The stated design-rule findings have not yet been resolved.

3. reserved

Designer: "The possible slot is not reserved."

The foundry has discussed an opportunity but has not allocated the slot.

4. approved

Liaison: "No sample-delivery date is approved."

The briefing supplies no approved commitment for finished samples.

08

Find the move you need

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

01 Wafer Fabrication Flow and Process Integration

Clarify the abbreviation: Here BEOL means back end of line.

Separate the stages: On-wafer interconnect formation is not package assembly.

State the hold: L24 is on hold pending inspection review.

Full phrase set: pages 7-8 | Dialogue: pages 10-11

02 Lithography, Reticles, and Critical Dimensions

Locate the dimension: Which layer and feature does the CD refer to?

Qualify the difference: The apparent difference is 3 nm.

Separate metrics: Overlay is not another name for critical dimension.

Full phrase set: pages 16-17 | Dialogue: pages 19-20

03 Deposition, Etch, CMP, and Process Windows

Identify the metric: The defined thickness-nonuniformity metric fell from 5% to 3%.

Separate changes: That is two percentage points, or a 40% relative reduction.

Ask about planarization: Has compatibility with the CMP step been assessed?

Full phrase set: pages 25-26 | Dialogue: pages 28-29

04 Metrology, SPC, and Yield Learning

Give the pilot counts: The pilot passed 92 of 100 tested die.

Name the relative basis: Relative to 90%, that is approximately 2.22%.

Request like-for-like data: Compare relevant product groups on a consistent basis.

Full phrase set: pages 34-35 | Dialogue: pages 37-38

Keep the conversation productive

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

05 Defect Density and Cleanroom Contamination

Replace the vague label: The record shows particle-like surface detections, not a confirmed cleanroom source.

Locate the pattern: The wafer map shows an edge cluster.

Separate measurement types: Surface detections are not airborne particle counts.

Full phrase set: pages 43-44 | Dialogue: pages 46-47

06 Equipment Uptime, Recipes, and Tool Matching

Confirm the recipe: Both tools used the identified R7 revision.

Give the difference: The means are 4 nm apart.

Ask about chamber state: Were the relevant chamber conditions documented?

Full phrase set: pages 52-53 | Dialogue: pages 55-56

07 Packaging, Test, and Reliability Qualification

Name the completed result: Q7 passed temperature cycling under the stated protocol.

Keep the plan visible: The approved plan requires all three tests and final review.

Ask about the vehicle: Is this the approved qualification vehicle?

Full phrase set: pages 61-62 | Dialogue: pages 64-65

08 Foundry, Tape-Out, PDK, and Capacity Communication

Name the requested milestone: You are requesting October 8 tape-out.

Confirm the basis: The checks use the agreed PDK revision P3.

Keep sign-off visible: The complete sign-off package is not yet available.

Full phrase set: pages 70-71 | Dialogue: pages 73-74

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

actual value compare setpoint and actual value	50
added defect quantify added defects	41
airborne particle count review airborne particle counts	42
ALD assess ALD conformality	23
alignment mark inspect alignment marks	15
anisotropy evaluate etch anisotropy	23
availability calculate equipment availability	50
backup tool qualify a backup tool	51
BEOL track BEOL progress	5
BGA assess BGA assembly	59
calibration status check calibration status	51
capacity reservation confirm capacity reservation	69
causal attribution qualify causal attribution	33
CD-SEM verify the CD-SEM recipe	14
chamber condition assess chamber condition	51
chamber matching verify chamber matching	50
CMP evaluate CMP compatibility	24
common-cause variation characterize common-cause variation	33
conformality assess film conformality	23
confounding factor identify a confounding factor	32
containment scope define containment scope	42
contamination source identify the contamination source	42
control chart review the control chart	33
control limit distinguish control limits	33

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

control signal respond to a control signal	33
corner condition evaluate corner conditions	24
corrective maintenance complete corrective maintenance	50
critical dimension measure critical dimension	14
cross-contamination prevent cross-contamination	42
CVD evaluate CVD film properties	23
cycle time state fabrication cycle time	69
data acceptance obtain foundry data acceptance	69
defect classification refine defect classification	41
defect density report defect density	41
defect review complete defect review	41
delamination inspect for delamination	60
delivery commitment qualify the delivery commitment	69
denominator state the denominator	32
deposition qualify a deposition step	23
depth of focus assess depth of focus	15
design rule check design rules	68
dicing complete wafer dicing	6
die identify the die	5
dielectric deposit a dielectric	6
dishing measure dishing	24
disposition obtain lot disposition	6
DRC resolve DRC findings	68
edge cluster investigate an edge cluster	41

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

electrical yield	report electrical yield	32
endpoint detection	verify endpoint detection	24
engineering change order	approve an engineering change order	69
engineering lot	plan an engineering lot	69
equipment state	report equipment state	50
erosion	assess erosion	24
etch bias	evaluate etch bias	14
etch selectivity	assess etch selectivity	23
exposure dose	record exposure dose	14
exposure history	reconstruct exposure history	42
exposure latitude	estimate exposure latitude	15
fab	move material through the fab	5
fabless company	support a fabless customer	68
failure analysis	initiate failure analysis	60
FEOL	complete FEOL processing	5
film stress	measure film stress	23
final test	complete final test	60
flip-chip attachment	evaluate flip-chip attachment	59
focus offset	evaluate focus offset	14
focus-exposure matrix	analyze a focus-exposure matrix	15
foundry	coordinate with the foundry	68
GDSII	deliver a GDSII dataset	68
HAST	schedule HAST	60
HTOL	report HTOL status	60

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

IDM work with an IDM	68
inspected area state the inspected area	41
inspection detection review inspection detections	41
inspection sensitivity confirm inspection sensitivity	42
integration trade-off explain the integration trade-off	24
interconnect form an interconnect	5
killer defect assess killer-defect relevance	41
line-edge roughness assess line-edge roughness	15
line-width roughness assess line-width roughness	15
lithography review lithography performance	14
lot identify the lot	6
lot hold place a lot on hold	6
LVS close an LVS mismatch	68
mask data preparation schedule mask data preparation	69
mask set release the mask set	69
matching criterion apply the matching criterion	51
measurement bias assess measurement bias	15
measurement recipe confirm the measurement recipe	14
measurement system assess the measurement system	33
metal layer identify the metal layer	5
metrology review metrology consistency	32
moisture sensitivity assess moisture sensitivity	59
MOL review MOL interfaces	5
mold compound qualify the mold compound	59

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

monitor wafer review monitor-wafer results	51
MTBF report MTBF	51
MTTR define the MTTR basis	51
nominal setting test the nominal setting	24
nuisance detection filter nuisance detections	41
OASIS verify the OASIS file	68
offset correction assess offset correction	51
optical proximity correction review optical proximity correction	15
overetch assess overetch effects	24
overlay measure overlay	15
package assembly begin package assembly	6
package construction identify package construction	59
package substrate specify the package substrate	59
particle excursion investigate a particle excursion	42
particle-like signature classify a particle-like signature	41
pass count preserve the pass count	32
pattern fidelity assess pattern fidelity	15
PDK confirm the PDK revision	68
percentage point report percentage-point change	32
photoresist coat the photoresist	14
pilot lot evaluate a pilot lot	32
pitch specify feature pitch	15
post-develop review post-develop CD	14
post-etch compare post-etch measurements	14

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

post-process scan	compare post-process scans	42
pre-process scan	preserve the pre-process scan	42
preconditioning	document preconditioning	60
precursor	qualify the precursor	23
preventive maintenance	schedule preventive maintenance	50
process flow	review the process flow	5
process integration	assess process integration	5
process interaction	investigate process interactions	24
process module	qualify a process module	6
process stability	assess process stability	33
process transfer	approve process transfer	51
process window	establish the process window	24
product mix	adjust for product mix	32
PVD	review PVD performance	23
QFN	review the QFN package	59
qualification approval	obtain qualification approval	60
qualification by similarity	assess qualification by similarity	60
qualification plan	follow the qualification plan	60
qualification vehicle	identify the qualification vehicle	60
qualification wafer	evaluate a qualification wafer	51
recipe release	obtain recipe release	24
recipe revision	confirm the recipe revision	50
reference lot	identify the reference lot	32
relative change	calculate relative change	32

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

release status confirm release status	51
remeasurement request comparable remeasurement	15
respin evaluate the need for a respin	69
reticle identify the reticle	14
risk production assess risk-production approval	69
route follow the approved route	6
rule deck identify the rule deck	68
sampling site align sampling sites	14
scheduled downtime plan scheduled downtime	50
seasoning verify seasoning status	51
setpoint record the setpoint	50
sign-off complete design sign-off	68
SiP evaluate a SiP design	59
size threshold state the size threshold	42
slurry qualify the CMP slurry	24
solder joint inspect solder joints	59
spatial signature compare spatial signatures	41
SPC apply SPC	33
special-cause variation investigate special-cause variation	33
specification limit check specification limits	33
step coverage evaluate step coverage	23
stratification stratify by product	32
subgroup define rational subgroups	33
surface contamination investigate surface contamination	42

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

suspect interval bound the suspect interval	42
tape-out plan the tape-out	68
test coverage assess qualification test coverage	60
tested population define the tested population	32
thermal cycling complete thermal cycling	60
thickness nonuniformity report thickness nonuniformity	23
thin film characterize the thin film	23
tool matching assess tool matching	50
traveler update the traveler	6
TSV characterize TSV connections	59
underfill assess underfill integrity	59
unscheduled downtime investigate unscheduled downtime	50
utilization state the utilization denominator	50
via inspect a via	5
wafer process a wafer	5
wafer fabrication track wafer fabrication	5
wafer ID confirm the wafer ID	6
wafer map examine the wafer map	41
wafer sort schedule wafer sort	6
wafer start confirm the wafer start	69
waiver request a defined waiver	69
WIP report WIP status	6
wire bonding qualify wire bonding	59
yield learning support yield learning	33

Original cases. Clear boundaries.

These are original fictional teaching conversations, not transcripts of real workers. Names, figures, organizations, and incidents are invented. Sources support terminology and communication aims, not the fictional results or an endorsement of this book.

[1] ASML. How Microchips Are Made.

[Open the original source](#)

Background on wafer processing, layers, lithography, and industry roles. The cases do not specify an actual fabrication flow or reproduce an equipment recipe. Checked 30 September 2026.

[2] NIST/SEMATECH. Engineering Statistics Handbook, Section 6.3.1: What Are Control Charts?

[Open the original source](#)

Background on monitoring process behavior. The yield comparisons use invented data and do not claim a completed statistical process study. Checked 30 September 2026.

[3] Texas Instruments. Reliability.

[Open the original source](#)

An industry example of reliability and qualification practice. The fictional three-test plan is not asserted as a universal qualification requirement. Checked 30 September 2026.

[4] SkyWater SKY130 PDK Documentation. Process Design Rules.

[Open the original source](#)

An open example of process-specific design rules, layer definitions, and physical verification terminology. The tape-out case does not use or approve a real design. Checked 30 September 2026.

Phil Stilwell

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